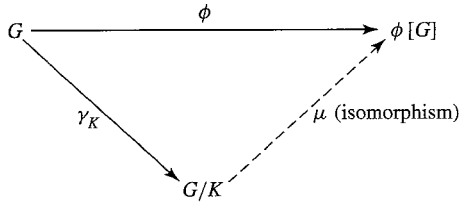
PART VII

Advanced Group Theory

Section 34 Isomorphism Theorems
Section 35 Series of Groups
Section 36 Sylow Theorems
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SECTION 34 ISOMORPHISM THEOREMS

There are several theorems concerning isomorphic factor groups that are known as the *isomorphism theorems* of group theory. The first of these is Theorem 14.11, which we restate for easy reference. The theorem is diagrammed in Fig. 34.1.



34.1 Figure

34.2 Theorem **(First Isomorphism Theorem)** Let $\phi : G \to G'$ be a homomorphism with kernel K, and let $\gamma_K : G \to G/K$ be the canonical homomorphism. There is a unique isomorphism $\mu : G/K \to \phi[G]$ such that $\phi(x) = \mu(\gamma_K(x))$ for each $x \in G$.

The lemma that follows will be of great aid in our proof and intuitive understanding of the other two isomorphism theorems.

34.3 Lemma Let N be a normal subgroup of a group G and let $\gamma : G \to G/N$ be the canonical homomorphism. Then the map ϕ from the set of normal subgroups of G containing N to the set of normal subgroups of G/N given by $\phi(L) = \gamma[L]$ is one to one and onto.

Proof Theorem 15.16 shows that if L is a normal subgroup of G containing N, then $\phi(L) = \gamma[L]$ is a normal subgroup of G/N. Because $N \leq L$, for each $x \in L$ the entire coset xN in G is contained in L. Thus by Theorem 13.15, $\gamma^{-1}[\phi(L)] = L$. Consequently, if L and M are normal subgroups of G, both containing N, and if $\phi(L) = \phi(M) = H$, then $L = \gamma^{-1}[H] = M$. Therefore ϕ is one to one.

If H is a normal subgroup of G/N, then $\gamma^{-1}[H]$ is a normal subgroup of G by Theorem 15.16. Because $N \in H$ and $\gamma^{-1}[\{N\}] = N$, we see that $N \subseteq \gamma^{-1}[H]$. Then $\phi(\gamma^{-1}[H]) = \gamma[\gamma^{-1}[H]] = H$. This shows that ϕ is onto the set of normal subgroups of G/N. ◆

If H and N are subgroups of a group G, then we let

$$HN = \{hn \mid h \in H, n \in N\}.$$

We define the **join** $H \vee N$ of H and N as the intersection of all subgroups of G that contain HN; thus $H \vee N$ is the smallest subgroup of G containing HN. Of course $H \vee N$ is also the smallest subgroup of G containing both H and N, since any such subgroup must contain HN. In general, HN need not be a subgroup of G. However, we have the following lemma.

34.4 Lemma If N is a normal subgroup of G, and if H is any subgroup of G, then $H \vee N = HN = NH$. Furthermore, if H is also normal in G, then HN is normal in G.

Proof We show that HN is a subgroup of G, from which $H \vee N = HN$ follows at once. Let $h_1, h_2 \in H$ and $n_1, n_2 \in N$. Since N is a normal subgroup, we have $n_1 h_2 = h_2 n_3$ for some $n_3 \in N$. Then $(h_1 n_1)(h_2 n_2) = h_1(n_1 h_2)n_2 = h_1(h_2 n_3)n_2 = (h_1 h_2)(n_3 n_2) \in HN$, so HN is closed under the induced operation in G. Clearly $e = ee$ is in HN. For $h \in H$ and $n \in N$, we have $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}n_4$ for some $n_4 \in N$, since N is a normal subgroup. Thus $(hn)^{-1} \in HN$, so $HN \leq G$. A similar argument shows that NH is a subgroup, so $NH = H \vee N = HN$.

Now suppose that H is also normal in G, and let $h \in H, n \in N$, and $g \in G$. Then $ghng^{-1} = (ghg^{-1})(gng^{-1}) \in HN$, so HN is indeed normal in G. ◆

We are now ready for the second isomorphism theorem.

34.5 Theorem **(Second Isomorphism Theorem)** Let H be a subgroup of G and let N be a normal subgroup of G. Then $(HN)/N \simeq H/(H \cap N)$.

Proof Let $\gamma : G \to G/N$ be the canonical homomorphism and let $H \leq G$. Then $\gamma[H]$ is a subgroup of G/N by Theorem 13.12. Now the action of γ on just the elements of H (called γ **restricted to** H) provides us with a homomorphism mapping H onto $\gamma[H]$, and the kernel of this restriction is clearly the set of elements of N that are also in H, that is, the intersection $H \cap N$. Theorem 34.2 then shows that there is an isomorphism $\mu_1 : H/(H \cap N) \to \gamma[H]$.

On the other hand, γ restricted to HN also provides a homomorphism mapping HN onto $\gamma[H]$, because $\gamma(n)$ is the identity N of G/N for all $n \in N$. The kernel of γ restricted to HN is N. Theorem 34.2 then provides us with an isomorphism $\mu_2 : (HN)/N \to \gamma[H]$.

Because $(HN)/N$ and $H/(H \cap N)$ are both isomorphic to $\gamma[H]$, they are isomorphic to each other. Indeed, $\phi : (HN)/N \to H/(H \cap N)$ where $\phi = {\mu_1}^{-1}\mu_2$ will be an isomorphism. More explicitly,

$$\phi((hn)N) = {\mu_1}^{-1}(\mu_2((hn)N)) = {\mu_1}^{-1}(h) = h(H \cap N). \quad \blacklozenge$$

34.6 Example Let $G = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$, $H = \mathbb{Z} \times \mathbb{Z} \times \{0\}$, and $N = \{0\} \times \mathbb{Z} \times \mathbb{Z}$. Then clearly $HN = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ and $H \cap N = \{0\} \times \mathbb{Z} \times \{0\}$. We have $(HN)/N \simeq \mathbb{Z}$ and we also have $H/(H \cap N) \simeq \mathbb{Z}$. ▲

If H and K are two normal subgroups of G and $K \leq H$, then H/K is a normal subgroup of G/K. The third isomorphism theorem concerns these groups.

34.7 Theorem **(Third Isomorphism Theorem)** Let H and K be normal subgroups of a group G with $K \leq H$. Then $G/H \simeq (G/K)/(H/K)$.

Proof Let $\phi : G \to (G/K)/(H/K)$ be given by $\phi(a) = (aK)(H/K)$ for $a \in G$. Clearly ϕ is onto $(G/K)/(H/K)$, and for $a, b \in G$,

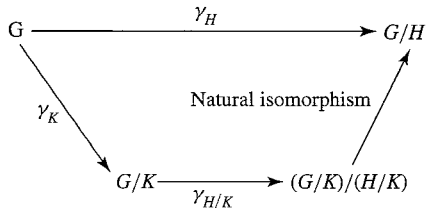
$$\begin{aligned}\phi(ab) &= [(ab)K](H/K) = [(aK)(bK)](H/K)\\ &= [(aK)(H/K)][(bK)(H/K)]\\ &= \phi(a)\phi(b),\end{aligned}$$

so ϕ is a homomorphism. The kernel consists of those $x \in G$ such that $\phi(x) = H/K$. These x are just the elements of H. Then Theorem 34.2 shows that $G/H \simeq (G/K)/(H/K)$. ◆

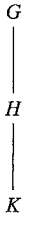
A nice way of viewing Theorem 34.7 is to regard the canonical map $\gamma_H : G \to G/H$ as being factored via a normal subgroup K of G, $K \leq H \leq G$, to give

$$\gamma_H = \gamma_{H/K}\gamma_K,$$

up to a natural isomorphism, as illustrated in Fig. 34.8. Another way of visualizing this theorem is to use the subgroup diagram in Fig. 34.9, where each group is a normal subgroup of G and is contained in the one above it. *The larger the normal subgroup, the smaller the factor group.* Thus we can think of G collapsed by H, that is, G/H, as being smaller than G collapsed by K. Theorem 34.7 states that we can collapse G all the way down to G/H in two steps. First, collapse to G/K, and then, using H/K, collapse this to $(G/K)/(H/K)$. The overall result is the same (up to isomorphism) as collapsing G by H.



34.8 Figure



34.9 Figure

34.10 Example Consider $K = 6\mathbb{Z} < H = 2\mathbb{Z} < G = \mathbb{Z}$. Then $G/H = \mathbb{Z}/2\mathbb{Z} \simeq \mathbb{Z}_2$. Now $G/K = \mathbb{Z}/6\mathbb{Z}$ has elements

$$6\mathbb{Z}, \quad 1+6\mathbb{Z}, \quad 2+6\mathbb{Z}, \quad 3+6\mathbb{Z}, \quad 4+6\mathbb{Z}, \quad \text{and} \quad 5+6\mathbb{Z}.$$

Of these six cosets, $6\mathbb{Z}$, $2+6\mathbb{Z}$, and $4+6\mathbb{Z}$ lie in $2\mathbb{Z}/6\mathbb{Z}$. Thus $(\mathbb{Z}/6\mathbb{Z})/(2\mathbb{Z}/6\mathbb{Z})$ has two elements and is isomorphic to $\mathbb{Z}_2$ also. Alternatively, we see that $\mathbb{Z}/6\mathbb{Z} \simeq \mathbb{Z}_6$, and $2\mathbb{Z}/6\mathbb{Z}$ corresponds *under this isomorphism* to the cyclic subgroup $\langle 2\rangle$ of $\mathbb{Z}_6$. Thus $(\mathbb{Z}/6\mathbb{Z})/(2\mathbb{Z}/6\mathbb{Z}) \simeq \mathbb{Z}_6/\langle 2\rangle \simeq \mathbb{Z}_2 \simeq \mathbb{Z}/2\mathbb{Z}$. ▲

■ EXERCISES 34

Computations

In using the three isomorphism theorems, it is often necessary to know the actual correspondence given by the isomorphism and not just the fact that the groups are isomorphic. The first six exercises give us training for this.

1. Let $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_3$ be the homomorphism such that $\phi(1) = 2$.
   - **a.** Find the kernel K of ϕ.
   - **b.** List the cosets in $\mathbb{Z}_{12}/K$, showing the elements in each coset.
   - **c.** Give the correspondence between $\mathbb{Z}_{12}/K$ and $\mathbb{Z}_3$ given by the map μ described in Theorem 34.2.
2. Let $\phi : \mathbb{Z}_{18} \to \mathbb{Z}_{12}$ be the homomorphism where $\phi(1) = 10$.
   - **a.** Find the kernel K of ϕ.
   - **b.** List the cosets in $\mathbb{Z}_{18}/K$, showing the elements in each coset.
   - **c.** Find the group $\phi[\mathbb{Z}_{18}]$.
   - **d.** Give the correspondence between $\mathbb{Z}_{18}/K$ and $\phi[\mathbb{Z}_{18}]$ given by the map μ described in Theorem 34.2.
3. In the group $\mathbb{Z}_{24}$, let $H = \langle 4\rangle$ and $N = \langle 6\rangle$.
   - **a.** List the elements in HN (which we might write $H + N$ for these additive groups) and in $H \cap N$.
   - **b.** List the cosets in HN/N, showing the elements in each coset.
   - **c.** List the cosets in $H/(H \cap N)$, showing the elements in each coset.
   - **d.** Give the correspondence between HN/N and $H/(H \cap N)$ described in the proof of Theorem 34.5.
4. Repeat Exercise 3 for the group $\mathbb{Z}_{36}$ with $H = \langle 6\rangle$ and $N = \langle 9\rangle$.
5. In the group $G = \mathbb{Z}_{24}$, let $H = \langle 4\rangle$ and $K = \langle 8\rangle$.
   - **a.** List the cosets in G/H, showing the elements in each coset.
   - **b.** List the cosets in G/K, showing the elements in each coset.
   - **c.** List the cosets in H/K, showing the elements in each coset.
   - **d.** List the cosets in $(G/K)/(H/K)$, showing the elements in each coset.
   - **e.** Give the correspondence between G/H and $(G/K)/(H/K)$ described in the proof of Theorem 34.7.
6. Repeat Exercise 5 for the group $G = \mathbb{Z}_{36}$ with $H = \langle 9\rangle$ and $K = \langle 18\rangle$.

Theory

7. Show directly from the definition of a normal subgroup that if H and N are subgroups of a group G, and N is normal in G, then $H \cap N$ is normal in H.

8. Let H, K, and L be normal subgroups of G with $H < K < L$. Let $A = G/H$, $B = K/H$, and $C = L/H$.

a. Show that B and C are normal subgroups of A, and $B < C$.

b. To what factor group of G is $(A/B)/(C/B)$ isomorphic?

9. Let K and L be normal subgroups of G with $K \vee L = G$, and $K \cap L = \{e\}$. Show that $G/K \simeq L$ and $G/L \simeq K$.

SECTION 35 SERIES OF GROUPS

Subnormal and Normal Series

This section is concerned with the notion of a *series* of a group G, which gives insight into the structure of G. The results hold for both abelian and nonabelian groups. They are not too important for finitely generated abelian groups because of our strong Theorem 11.12. Many of our illustrations will be taken from abelian groups, however, for ease of computation.

35.1 Definition A **subnormal** (or **subinvariant**) **series of a group** G is a finite sequence $H_0, H_1, \cdots, H_n$ of subgroups of G such that $H_i < H_{i+1}$ and H_i is a normal subgroup of H_{i+1} with $H_0 = \{e\}$ and $H_n = G$. A **normal** (or **invariant**) **series of** G is a finite sequence $H_0, H_1, \cdots, H_n$ of normal subgroups of G such that $H_i < H_{i+1}$, $H_0 = \{e\}$, and $H_n = G$. ■

Note that for abelian groups the notions of subnormal and normal series coincide, since every subgroup is normal. A normal series is always subnormal, but the converse need not be true. We defined a subnormal series before a normal series, since the concept of a subnormal series is more important for our work.

35.2 Example Two examples of normal series of $\mathbb{Z}$ under addition are

$$\{0\} < 8\mathbb{Z} < 4\mathbb{Z} < \mathbb{Z}$$

and

$$\{0\} < 9\mathbb{Z} < \mathbb{Z}.$$

▲

35.3 Example Consider the group D_4 of symmetries of the square in Example 8.10. The series

$$\{\rho_0\} < \{\rho_0, \mu_1\} < \{\rho_0, \rho_2, \mu_1, \mu_2\} < D_4$$

is a subnormal series, as we could check using Table 8.12. It is not a normal series since $\{\rho_0, \mu_1\}$ is not normal in D_4. ▲

35.4 Definition A subnormal (normal) series $\{K_j\}$ is a **refinement of a subnormal (normal) series** $\{H_i\}$ of a group G if $\{H_i\} \subseteq \{K_j\}$, that is, if each H_i is one of the K_j. ■

35.5 Example The series

$$\{0\} < 72\mathbb{Z} < 24\mathbb{Z} < 8\mathbb{Z} < 4\mathbb{Z} < \mathbb{Z}$$

is a refinement of the series

$$\{0\} < 72\mathbb{Z} < 8\mathbb{Z} < \mathbb{Z}.$$

Two new terms, $4\mathbb{Z}$ and $24\mathbb{Z}$, have been inserted. ▲

Of interest in studying the structure of G are the factor groups H_{i+1}/H_i. These are defined for both normal and subnormal series, since H_i is normal in H_{i+1} in either case.

35.6 Definition Two subnormal (normal) series $\{H_i\}$ and $\{K_j\}$ of the same group G are **isomorphic** if there is a one-to-one correspondence between the collections of factor groups $\{H_{i+1}/H_i\}$ and $\{K_{j+1}/K_j\}$ such that corresponding factor groups are isomorphic. ■

Clearly, two isomorphic subnormal (normal) series must have the same number of groups.

35.7 Example The two series of $\mathbb{Z}_{15}$,

$$\{0\} < \langle 5\rangle < \mathbb{Z}_{15}$$

and

$$\{0\} < \langle 3\rangle < \mathbb{Z}_{15},$$

are isomorphic. Both $\mathbb{Z}_{15}/\langle 5\rangle$ and $\langle 3\rangle/\{0\}$ are isomorphic to $\mathbb{Z}_5$, and $\mathbb{Z}_{15}/\langle 3\rangle$ is isomorphic to $\langle 5\rangle/\{0\}$, or to $\mathbb{Z}_3$. ▲

The Schreier Theorem

We proceed to prove that two subnormal series of a group G have isomorphic refinements. This is a fundamental result in the theory of series. The proof is not too difficult. However, we know from experience that some students get lost in the proof, and then tend to feel that they cannot understand the theorem. We now give an illustration of the theorem before we proceed to its proof.

35.8 Example Let us try to find isomorphic refinements of the series

$$\{0\} < 8\mathbb{Z} < 4\mathbb{Z} < \mathbb{Z}$$

and

$$\{0\} < 9\mathbb{Z} < \mathbb{Z}$$

given in Example 35.2. Consider the refinement

$$\{0\} < 72\mathbb{Z} < 8\mathbb{Z} < 4\mathbb{Z} < \mathbb{Z}$$

of $\{0\} < 8\mathbb{Z} < 4\mathbb{Z} < \mathbb{Z}$ and the refinement

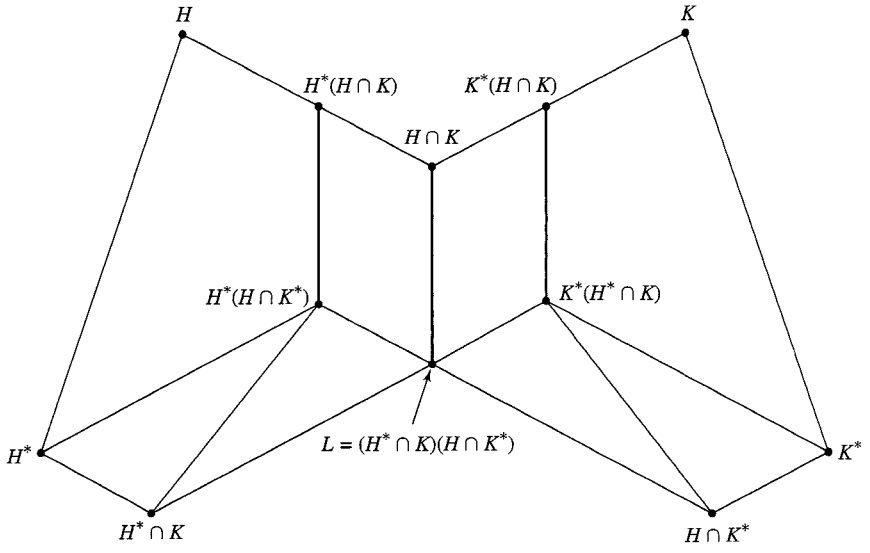
$$\{0\} < 72\mathbb{Z} < 18\mathbb{Z} < 9\mathbb{Z} < \mathbb{Z}$$

of $\{0\} < 9\mathbb{Z} < \mathbb{Z}$. In both cases the refinements have four factor groups isomorphic to $\mathbb{Z}_4, \mathbb{Z}_2, \mathbb{Z}_9$, and $72\mathbb{Z}$ or $\mathbb{Z}$. The *order* in which the factor groups occur is different to be sure. ▲

We start with a rather technical lemma developed by Zassenhaus. This lemma is sometimes called the *butterfly lemma*, since Fig. 35.9, which accompanies the lemma, has a butterfly shape.

Let H and K be subgroups of a group G, and let H^* be a normal subgroup of H and K^* be a normal subgroup of K. Applying the first statement in Lemma 34.4 to H^* and $H \cap K$ as subgroups of H, we see that $H^*(H \cap K)$ is a group. Similar arguments show that $H^*(H \cap K^*)$, $K^*(H \cap K)$, and $K^*(H^* \cap K)$ are also groups. It is not hard to show that $H^* \cap K$ is a normal subgroup of $H \cap K$ (see Exercise 22). The same argument using Lemma 34.4 applied to $H^* \cap K$ and $H \cap K^*$ as subgroups of $H \cap K$ shows that $L = (H^* \cap K)(H \cap K^*)$ is a group. Thus we have the diagram of subgroups shown in Fig. 35.9. It is not hard to verify the inclusion relations indicated by the diagram.

Since both $H \cap K^*$ and $H^* \cap K$ are normal subgroups of $H \cap K$, the second statement in Lemma 34.4 shows that $L = (H^* \cap K)(H \cap K^*)$ is a normal subgroup of $H \cap K$. We have denoted this particular normal subgroup relationship by the heavy middle line in Fig. 35.9. We claim the other two heavy lines also indicate normal subgroup relationships, and that the three factor groups given by the three normal subgroup relations are all isomorphic. To show this, we shall define a homomorphism $\phi : H^*(H \cap K) \to (H \cap K)/L$, and show that ϕ is onto $(H \cap K)/L$ with kernel $H^*(H \cap K^*)$. It will then follow at once from Theorem 34.2 that $H^*(H \cap K^*)$ is normal



35.9 Figure

in $H^*(H \cap K)$, and that $H^*(H \cap K)/H^*(H \cap K^*) \simeq (H \cap K)/L$. A similar result for the groups on the right-hand heavy line in Fig. 35.9 then follows by symmetry.

Let $\phi : H^*(H \cap K) \to (H \cap K)/L$ be defined as follows. For $h \in H^*$ and $x \in H \cap K$, let $\phi(hx) = xL$. We show ϕ is well-defined and a homomorphism. Let $h_1, h_2 \in H^*$ and $x_1, x_2 \in H \cap K$. If $h_1x_1 = h_2x_2$, then $h_2^{-1}h_1 = x_2x_1^{-1} \in H^* \cap (H \cap K) = H^* \cap K \subseteq L$, so $x_1L = x_2L$. Thus ϕ is well defined. Since H^* is normal in H, there is h_3 in H^* such that $x_1h_2 = h_3x_1$. Then

$$\begin{aligned}\phi((h_1x_1)(h_2x_2)) &= \phi((h_1h_3)(x_1x_2)) = (x_1x_2)L \\ &= (x_1L)(x_2L) = \phi(h_1x_1) \cdot \phi(h_2x_2).\end{aligned}$$

Thus ϕ is a homomorphism.

Obviously ϕ is onto $(H \cap K)/L$. Finally if $h \in H^*$ and $x \in H \cap K$, then $\phi(Hx) = xL = L$ if and only if $x \in L$, or if and only if $hx \in H^*L = H^*(H^* \cap K)(H \cap K^*) = H^*(H \cap K^*)$. Thus $\text{Ker}(\phi) = H^*(H \cap K^*)$.

We have proved the following lemma.

35.10 Lemma **(Zassenhaus Lemma)** Let H and K be subgroups of a group G and let H^* and K^* be normal subgroups of H and K, respectively. Then

1. $H^*(H \cap K^*)$ is a normal subgroup of $H^*(H \cap K)$.
2. $K^*(H^* \cap K)$ is a normal subgroup of $K^*(H \cap K)$.
3. $$\begin{aligned}H^*(H \cap K)/H^*(H \cap K^*) &\simeq K^*(H \cap K)/K^*(H^* \cap K) \\ &\simeq (H \cap K)/[(H^* \cap K)(H \cap K^*)].\end{aligned}$$

35.11 Theorem **(Schreier Theorem)** Two subnormal (normal) series of a group G have isomorphic refinements.

Proof Let G be a group and let

$$\{e\} = H_0 < H_1 < H_2 < \cdots < H_n = G \tag{1}$$

and

$$\{e\} = K_0 < K_1 < K_2 < \cdots < K_m = G \tag{2}$$

be two subnormal series for G. For i where $0 \leq i \leq n-1$, form the chain of groups

$$H_i = H_i(H_{i+1} \cap K_0) \leq H_i(H_{i+1} \cap K_1) \leq \cdots \leq H_i(H_{i+1} \cap K_m) = H_{i+1}.$$

This inserts $m-1$ not necessarily distinct groups between H_i and H_{i+1}. If we do this for each i where $0 \leq i \leq n-1$ and let $H_{ij} = H_i(H_{i+1} \cap K_j)$, then we obtain the chain of groups

$$\begin{aligned}\{e\} = H_{0,0} &\leq H_{0,1} \leq H_{0,2} \leq \cdots \leq H_{0,m-1} \leq H_{1,0} \\ &\leq H_{1,1} \leq H_{1,2} \leq \cdots \leq H_{1,m-1} \leq H_{2,0} \\ &\leq H_{2,1} \leq H_{2,2} \leq \cdots \leq H_{2,m-1} \leq H_{3,0} \\ &\leq \cdots \\ &\leq H_{n-1,1} \leq H_{n-1,2} \leq \cdots \leq H_{n-1,m-1} \leq H_{n-1,m} \\ &= G.\end{aligned} \tag{3}$$

This chain (3) contains $nm + 1$ not necessarily distinct groups, and $H_{i,0} = H_i$ for each i. By the Zassenhaus lemma, chain (3) is a subnormal chain, that is, each group is normal in the following group. This chain refines the series (1).

In a symmetric fashion, we set $K_{j,i} = K_j(K_{j+1} \cap H_i)$ for $0 \leq j \leq m-1$ and $0 \leq i \leq n$. This gives a subnormal chain

$$\begin{aligned}\{e\} = K_{0,0} &\leq K_{0,1} \leq K_{0,2} \leq \cdots \leq K_{0,n-1} \leq K_{1,0} \\ &\leq K_{1,1} \leq K_{1,2} \leq \cdots \leq K_{1,n-1} \leq K_{2,0} \\ &\leq K_{2,1} \leq K_{2,2} \leq \cdots \leq K_{2,n-1} \leq K_{3,0} \\ &\leq \cdots \\ &\leq K_{m-1,1} \leq K_{m-1,2} \leq \cdots \leq K_{m-1,n-1} \leq K_{m-1,n} \\ &= G. \end{aligned} \tag{4}$$

This chain (4) contains $mn + 1$ not necessarily distinct groups, and $K_{j,0} = K_j$ for each j. This chain refines the series (2).

By the Zassenhaus lemma 35.10, we have

$$H_i(H_{i+1} \cap K_{j+1})/H_i(H_{i+1} \cap K_j) \simeq K_j(K_{j+1} \cap H_{i+1})/K_j(K_{j+1} \cap H_i),$$

or

$$H_{i,j+1}/H_{i,j} \simeq K_{j,i+1}/K_{j,i} \tag{5}$$

for $0 \leq i \leq n-1$ and $0 \leq j \leq m-1$. The isomorphisms of relation (5) give a one-to-one correspondence of isomorphic factor groups between the subnormal chains (3) and (4). To verify this correspondence, note that $H_{i,0} = H_i$ and $H_{i,m} = H_{i+1}$, while $K_{j,0} = K_j$ and $K_{j,n} = K_{j+1}$. Each chain in (3) and (4) contains a rectangular array of mn symbols $\leq$. Each $\leq$ gives rise to a factor group. The factor groups arising from the rth *row* of $\leq$'s in chain (3) correspond to the factor groups arising from the rth *column* of $\leq$'s in chain (4). Deleting repeated groups from the chains in (3) and (4), we obtain subnormal series of distinct groups that are isomorphic refinements of chains (1) and (2). This establishes the theorem for subnormal series.

For normal series, where all H_i and K_j are normal in G, we merely observe that all the groups $H_{i,j}$ and $K_{j,i}$ formed above are also normal in G, so the same proof applies. This normality of $H_{i,j}$ and $K_{j,i}$ follows at once from the second assertion in Lemma 34.4 and from the fact that intersections of normal subgroups of a group yield normal subgroups. ◆

The Jordan–Hölder Theorem

We now come to the real meat of the theory.

35.12 Definition A subnormal series $\{H_i\}$ of a group G is a **composition series** if all the factor groups H_{i+1}/H_i are simple. A normal series $\{H_i\}$ of G is a **principal** or **chief series** if all the factor groups H_{i+1}/H_i are simple. ■

Note that for abelian groups the concepts of composition and principal series coincide. Also, since every normal series is subnormal, every principal series is a composition series for any group, abelian or not.

35.13 Example We claim that $\mathbb{Z}$ has no composition (and also no principal) series. For if

$$\{0\} = H_0 < H_1 < \cdots < H_{n-1} < H_n = \mathbb{Z}$$

is a subnormal series, H_1 must be of the form $r\mathbb{Z}$ for some $r \in \mathbb{Z}^+$. But then H_1/H_0 is isomorphic to $r\mathbb{Z}$, which is infinite cyclic with many nontrivial proper normal subgroups, for example, $2r\mathbb{Z}$. Thus $\mathbb{Z}$ has no composition (and also no principal) series. ▲

35.14 Example The series

$$\{e\} < A_n < S_n$$

for $n \geq 5$ is a composition series (and also a principal series) of S_n, because $A_n/\{e\}$ is isomorphic to A_n, which is simple for $n \geq 5$, and S_n/A_n is isomorphic to $\mathbb{Z}_2$, which is simple. Likewise, the two series given in Example 35.7 are composition series (and also principal series) of $\mathbb{Z}_{15}$. They are isomorphic, as shown in that example. This illustrates our main theorem, which will be stated shortly. ▲

Observe that by Theorem 15.18, H_{i+1}/H_i is simple if and only if H_i is a maximal normal subgroup of H_{i+1}. Thus for a composition series, each H_i must be a maximal normal subgroup of H_{i+1}. *To form a composition series of a group G, we just hunt for a maximal normal subgroup H_{n-1} of G, then for a maximal normal subgroup H_{n-2} of H_{n-1}, and so on. If this process terminates in a finite number of steps, we have a composition series.* Note that by Theorem 15.18, a composition series cannot have any further refinement. *To form a principal series, we have to hunt for a maximal normal subgroup H_{n-1} of G, then for a maximal normal subgroup H_{n-2} of H_{n-1} that is also normal in G, and so on.* The main theorem is as follows.

35.15 Theorem **(Jordan–Hölder Theorem)** Any two composition (principal) series of a group G are isomorphic.

Proof Let $\{H_i\}$ and $\{K_i\}$ be two composition (principal) series of G. By Theorem 35.11, they have isomorphic refinements. But since all factor groups are already simple, Theorem 15.18 shows that neither series has any further refinement. Thus $\{H_i\}$ and $\{K_i\}$ must already be isomorphic. ◆

For a finite group, we should regard a composition series as a type of factorization of the group into simple factor groups, analogous to the factorization of a positive integer into primes. In both cases, the factorization is unique, up to the order of the factors.

■ Historical Note

This first appearance of what became the Jordan–Hölder theorem occurred in 1869 in a commentary on the work of Galois by the brilliant French algebraist Camille Jordan (1838–1922). The context of its appearance is the study of permutation groups associated with the roots of polynomial equations. Jordan asserted that even though the sequence of normal subgroups $G, I, J, \cdots$ of the group of the equation is not necessarily unique, nevertheless the sequence of indices of this composition series is unique. Jordan gave a proof in his monumental 1870 *Treatise on Substitutions and Algebraic Equations*. This latter work, though restricted to what we now call permutation groups, remained the standard treatise on group theory for many years.

The Hölder part of the theorem, that the sequence of factor groups in a composition series is unique up to order, was due to Otto Hölder (1859–1937), who played a very important role in the development of group theory once the completely abstract definition of a group had been given. Among his other contributions, he gave the first abstract definition of a "factor group" and determined the structure of all finite groups of square-free order.

35.16 Theorem If G has a composition (principal) series, and if N is a proper normal subgroup of G, then there exists a composition (principal) series containing N.

Proof The series

$$\{e\} < N < G$$

is both a subnormal and a normal series. Since G has a composition series $\{H_i\}$, then by Theorem 35.11 there is a refinement of $\{e\} < N < G$ to a subnormal series isomorphic to a refinement of $\{H_i\}$. But as a composition series, $\{H_i\}$ can have no further refinement. Thus $\{e\} < N < G$ can be refined to a subnormal series all of whose factor groups are simple, that is, to a composition series. A similar argument holds if we start with a principal series $\{K_j\}$ of G. ◆

35.17 Example A composition (and also a principal) series of $\mathbb{Z}_4 \times \mathbb{Z}_9$ containing $\langle(0, 1)\rangle$ is

$$\{(0, 0)\} < \langle(0, 3)\rangle < \langle(0, 1)\rangle < \langle 2\rangle \times \langle 1\rangle < \langle 1\rangle \times \langle 1\rangle = \mathbb{Z}_4 \times \mathbb{Z}_9. \quad ▲$$

The next definition is basic to the characterization of those polynomial equations whose solutions can be expressed in terms of radicals.

35.18 Definition A group G is **solvable** if it has a composition series $\{H_i\}$ such that all factor groups H_{i+1}/H_i are abelian. ■

By the Jordan–Hölder theorem, we see that for a solvable group, *every* composition series $\{H_i\}$ must have abelian factor groups H_{i+1}/H_i.

35.19 Example The group S_3 is solvable, because the composition series

$$\{e\} < A_3 < S_3$$

has factor groups isomorphic to $\mathbb{Z}_3$ and $\mathbb{Z}_2$, which are abelian. The group S_5 is not solvable, for since A_5 is simple, the series

$$\{e\} < A_5 < S_5$$

is a composition series, and $A_5/\{e\}$, which is isomorphic to A_5, is not abelian. *This group A_5 of order 60 can be shown to be the smallest group that is not solvable.* This fact is closely connected with the fact that a polynomial equation of degree 5 is not in general solvable by radicals, but a polynomial equation of degree ≤ 4 is. ▲

The Ascending Central Series

We mention one subnormal series for a group G that can be formed using centers of groups. Recall from Section 15 that the center $Z(G)$ of a group G is defined by

$$Z(G) = \{z \in G \mid zg = gz \text{ for all } g \in G\},$$

and that $Z(G)$ is a normal subgroup of G. If we have the table for a finite group G, it is easy to find the center. An element a will be in the center of G if and only if the elements in the row opposite a at the extreme left are given in the same order as the elements in the column under a at the very top of the table.

Now let G be a group, and let $Z(G)$ be the center of G. Since $Z(G)$ is normal in G, we can form the factor group $G/Z(G)$ and find the center $Z(G/Z(G))$ of this factor group. Since $Z(G/Z(G))$ is normal in $G/Z(G)$, if $\gamma : G \to G/Z(G)$ is the canonical map, then by Theorem 15.16, $\gamma^{-1}[Z(G/Z(G))]$ is a normal subgroup $Z_1(G)$ of G. We can then form the factor group $G/Z_1(G)$ and find its center, take $(\gamma_1)^{-1}$ of it to get $Z_2(G)$, and so on.

35.20 Definition The series

$$\{e\} \leq Z(G) \leq Z_1(G) \leq Z_2(G) \leq \cdots$$

described in the preceding discussion is the **ascending central series of the group G.** ■

35.21 Example The center of S_3 is just the identity $\{\rho_0\}$. Thus the ascending central series of S_3 is

$$\{\rho_0\} \leq \{\rho_0\} \leq \{\rho_0\} \leq \cdots.$$

The center of the group D_4 of symmetries of the square in Example 8.10 is $\{\rho_0, \rho_2\}$. (Do you remember that we said that this group would give us nice examples of many things we discussed?) Since $D_4/\{\rho_0, \rho_2\}$ is of order 4 and hence abelian, its center is all of $D_4/\{\rho_0, \rho_2\}$. Thus the ascending central series of D_4 is

$$\{\rho_0\} \leq \{\rho_0, \rho_2\} \leq D_4 \leq D_4 \leq D_4 \leq \cdots. \quad ▲$$

EXERCISES 35

Computations

In Exercises 1 through 5, give isomorphic refinements of the two series.

1. $\{0\} < 10\mathbb{Z} < \mathbb{Z}$ and $\{0\} < 25\mathbb{Z} < \mathbb{Z}$

2. $\{0\} < 60\mathbb{Z} < 20\mathbb{Z} < \mathbb{Z}$ and $\{0\} < 245\mathbb{Z} < 49\mathbb{Z} < \mathbb{Z}$

3. $\{0\} < \langle 3\rangle < \mathbb{Z}_{24}$ and $\{0\} < \langle 8\rangle < \mathbb{Z}_{24}$

4. $\{0\} < \langle 18\rangle < \langle 3\rangle < \mathbb{Z}_{72}$ and $\{0\} < \langle 24\rangle < \langle 12\rangle < \mathbb{Z}_{72}$

5. $\{(0,0)\} < (60\mathbb{Z}) \times \mathbb{Z} < (10\mathbb{Z}) \times \mathbb{Z} < \mathbb{Z} \times \mathbb{Z}$ and $\{(0,0)\} < \mathbb{Z} \times (80\mathbb{Z}) < \mathbb{Z} \times (20\mathbb{Z}) < \mathbb{Z} \times \mathbb{Z}$

6. Find all composition series of $\mathbb{Z}_{60}$ and show that they are isomorphic.

7. Find all composition series of $\mathbb{Z}_{48}$ and show that they are isomorphic.

8. Find all composition series of $\mathbb{Z}_5 \times \mathbb{Z}_5$.

9. Find all composition series of $S_3 \times \mathbb{Z}_2$.

10. Find all composition series of $\mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$.

11. Find the center of $S_3 \times \mathbb{Z}_4$.

12. Find the center of $S_3 \times D_4$.

13. Find the ascending central series of $S_3 \times \mathbb{Z}_4$.

14. Find the ascending central series of $S_3 \times D_4$.

Concepts

In Exercises 15 and 16, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

15. A *composition series* of a group G is a finite sequence

$$\{e\} = H_0 < H_1 < H_2 < \cdots < H_{n-1} < H_n = G$$

of subgroups of G such that H_i is a maximal normal subgroup of H_{i+1} for $i = 0, 1, 2, \cdots, n-1$.

16. A *solvable group* is one that has a composition series of abelian groups.

17. Mark each of the following true or false.

_____ **a.** Every normal series is also subnormal.

_____ **b.** Every subnormal series is also normal.

_____ **c.** Every principal series is a composition series.

_____ **d.** Every composition series is a principal series.

_____ **e.** Every abelian group has exactly one composition series.

_____ **f.** Every finite group has a composition series.

_____ **g.** A group is solvable if and only if it has a composition series with simple factor groups.

_____ **h.** S_7 is a solvable group.

_____ **i.** The Jordan–Hölder theorem has some similarity with the Fundamental Theorem of Arithmetic, which states that every positive integer greater than 1 can be factored into a product of primes uniquely up to order.

_____ **j.** Every finite group of prime order is solvable.

18. Find a composition series of $S_3 \times S_3$. Is $S_3 \times S_3$ solvable?

19. Is the group D_4 of symmetries of the square in Example 8.10 solvable?

20. Let G be $\mathbb{Z}_{36}$. Refer to the proof of Theorem 35.11. Let the subnormal series (1) be

$$\{0\} < \langle 12 \rangle < \langle 3 \rangle < \mathbb{Z}_{36}$$

and let the subnormal series (2) be

$$\{0\} < \langle 18 \rangle < \mathbb{Z}_{36}.$$

Find chains (3) and (4) and exhibit the isomorphic factor groups as described in the proof. Write chains (3) and (4) in the rectangular array shown in the text.

21. Repeat Exercise 20 for the group $\mathbb{Z}_{24}$ with the subnormal series (1)

$$\{0\} < \langle 12 \rangle < \langle 4 \rangle < \mathbb{Z}_{24}$$

and (2)

$$\{0\} < \langle 6 \rangle < \langle 3 \rangle < \mathbb{Z}_{24}.$$

Theory

22. Let H^*, H, and K be subgroups of G with H^* normal in H. Show that $H^* \cap K$ is normal in $H \cap K$.

23. Show that if

$$H_0 = \{e\} < H_1 < H_2 < \cdots < H_n = G$$

is a subnormal (normal) series for a group G, and if H_{i+1}/H_i is of finite order s_{i+1}, then G is of finite order $s_1 s_2 \cdots s_n$.

24. Show that an infinite abelian group can have no composition series. [*Hint:* Use Exercise 23, together with the fact that an infinite abelian group always has a proper normal subgroup.]

25. Show that a finite direct product of solvable groups is solvable.

26. Show that a subgroup K of a solvable group G is solvable. [*Hint:* Let $H_0 = \{e\} < H_1 < \cdots < H_n = G$ be a composition series for G. Show that the distinct groups among $K \cap H_i$ for $i = 0, \cdots, n$ form a composition series for K. Observe that

$$(K \cap H_i)/(K \cap H_{i-1}) \simeq [H_{i-1}(K \cap H_i)]/[H_{i-1}],$$

by Theorem 34.5, with $H = K \cap H_i$ and $N = H_{i-1}$, and that $H_{i-1}(K \cap H_i) \leq H_i$.]

27. Let $H_0 = \{e\} < H_1 < \cdots < H_n = G$ be a composition series for a group G. Let N be a normal subgroup of G, and suppose that N is a simple group. Show that the distinct groups among H_0, $H_i N$ for $i = 0, \cdots, n$ also form a composition series for G. [*Hint:* $H_i N$ is a group by Lemma 34.4. Show that $H_{i-1}N$ is normal in $H_i N$. By Theorem 34.5

$$(H_i N)/(H_{i-1}N) \simeq H_i/[H_i \cap (H_{i-1}N)],$$

and the latter group is isomorphic to

$$[H_i/H_{i-1}]/[(H_i \cap (H_{i-1}N))/H_{i-1}],$$

by Theorem 34.7. But H_i/H_{i-1} is simple.]

28. Let G be a group, and let $H_0 = \{e\} < H_1 < \cdots < H_n = G$ be a composition series for G. Let N be a normal subgroup of G, and let $\gamma : G \to G/N$ be the canonical map. Show that the distinct groups among $\gamma[H_i]$ for $i = 0, \cdots, n$, form a composition series for G/N. [*Hint:* Observe that the map

$$\psi : H_iN \to \gamma(H_i)/\gamma[H_{i-1}]$$

defined by

$$\psi(h_in) = \gamma(h_in)\gamma[H_{i-1}]$$

is a homomorphism with kernel $H_{i-1}N$. By Theorem 34.2.

$$\gamma(H_i)/\gamma[H_{i-1}] \simeq (H_iN)/(H_{i-1}N).$$

Proceed via Theorem 34.5, as shown in the hint for Exercise 27.]

29. Prove that a homomorphic image of a solvable group is solvable. [*Hint:* Apply Exercise 28 to get a composition series for the homomorphic image. The hints for Exercises 27 and 28 then show how the factor groups of this composition series in the image look.]

SECTION 36 SYLOW THEOREMS

The fundamental theorem for finitely generated abelian groups (Theorem 11.12) gives us complete information about all finite abelian groups. The study of finite nonabelian groups is much more complicated. The Sylow theorems give us some important information about them.

We know the order of a subgroup of a finite group G must divide $|G|$. If G is abelian, then there exist subgroups of every order dividing $|G|$. We showed in Example 15.6 that A_4, which has order 12, has no subgroup of order 6. Thus a nonabelian group G may have no subgroup of some order d dividing $|G|$; the "converse of the theorem of Lagrange" does not hold. The Sylow theorems give a weak converse. Namely, they show that if d is a power of a prime and d divides $|G|$, then G does contain a subgroup of order d. (Note that 6 is not a power of a prime.) The Sylow theorems also give some information concerning the number of such subgroups and their relationship to each other. We will see that these theorems are very useful in studying finite nonabelian groups.

Proofs of the Sylow theorems give us another application of action of a group on a set described in Section 16. This time, the set itself is formed from the group; in some instances the set is the group itself, sometimes it is a collection of cosets of a subgroup, and sometimes it is a collection of subgroups.

p-Groups

Section 17 gave applications of Burnside's formula that counted the number of orbits in a finite G-set. Most of our results in this section flow from an equation that counts the number of elements in a finite G-set.

Let X be a finite G-set. Recall that for $x \in X$, the orbit of x in X under G is $Gx = \{gx \mid g \in G\}$. Suppose that there are r orbits in X under G, and let $\{x_1, x_2, \cdots, x_r\}$ contain one element from each orbit in X. Now every element of X is in precisely one

orbit, so

$$|X| = \sum_{i=1}^{r} |Gx_i|. \tag{1}$$

There may be one-element orbits in X. Let $X_G = \{x \in X \mid gx = x \text{ for all } g \in G\}$. Thus X_G is precisely the union of the one-element orbits in X. Let us suppose there are s one-element orbits, where $0 \leq s \leq r$. Then $|X_G| = s$, and reordering the x_i if necessary, we may rewrite Eq. (1) as

$$|X| = |X_G| + \sum_{i=s+1}^{r} |Gx_i|. \tag{2}$$

Most of the results of this section will flow from Eq. (2). We shall develop Sylow theory as in Hungerford [10], where credit is given to R. J. Nunke for the line of proof. The proof of Theorem 36.3 (Cauchy's theorem) is credited there to J. H. McKay.

Theorem 36.1, which follows, is not quite a counting theorem, but it does have a numerical conclusion. It counts modulo p. The theorem seems to be amazingly powerful. In the rest of the chapter, if we choose the correct set, the correct group action on it, and apply Theorem 36.1, what we want seems to fall right into our lap! Compared with older proofs, the arguments are extremely pretty and elegant.

Throughout this section, p will always be a prime integer.

36.1 Theorem Let G be a group of order p^n and let X be a finite G-set. Then $|X| \equiv |X_G| \pmod{p}$.

Proof In the notation of Eq. (2), we know that $|Gx_i|$ divides $|G|$ by Theorem 16.16. Consequently p divides $|Gx_i|$ for $s + 1 \leq i \leq r$. Equation (2) then shows that $|X| - |X_G|$ is divisible by p, so $|X| \equiv |X_G| \pmod{p}$. ◆

36.2 Definition Let p be a prime. A group G is a p-**group** if every element in G has order a power of the prime p. A subgroup of a group G is a p-**subgroup of** G if the subgroup is itself a p-group. ■

Our goal in this section is to show that a finite group G has a subgroup of every prime-power order dividing $|G|$. As a first step, we prove Cauchy's theorem, which says that if p divides $|G|$, then G has a subgroup of order p.

36.3 Theorem **(Cauchy's Theorem)** Let p be a prime. Let G be a finite group and let p divide $|G|$. Then G has an element of order p and, consequently, a subgroup of order p.

Proof We form the set X of all p-tuples $(g_1, g_2, \cdots, g_p)$ of elements of G having the property that the product of the coordinates in G is e. That is,

$$X = \{(g_1, g_2, \cdots, g_p) \mid g_i \in G \text{ and } g_1 g_2 \cdots g_p = e\}.$$

We claim p divides $|X|$. In forming a p-tuple in X, we may let $g_1, g_2, \cdots, g_{p-1}$ be any elements of G, and g_p is then uniquely determined as $(g_1 g_2 \cdots g_{p-1})^{-1}$. Thus $|X| = |G|^{p-1}$ and since p divides $|G|$, we see that p divides $|X|$.

Let σ be the cycle $(1, 2, 3, \cdots, p)$ in S_p. We let σ act on X by

$$\sigma(g_1, g_2, \cdots, g_p) = (g_{\sigma(1)}, g_{\sigma(2)}, \cdots, g_{\sigma(p)}) = (g_2, g_3, \cdots, g_p, g_1).$$

Note that $(g_2, g_3, \cdots, g_p, g_1) \in X$, for $g_1(g_2 g_3 \cdots g_p) = e$ implies that $g_1 = (g_2 g_3 \cdots g_p)^{-1}$, so $(g_2 g_3 \cdots g_p) g_1 = e$ also. Thus σ acts on X, and we consider the subgroup $\langle \sigma \rangle$ of S_p to act on X by iteration in the natural way.

Now $|\langle \sigma \rangle| = p$, so we may apply Theorem 36.1, and we know that $|X| \equiv |X_{\langle \sigma \rangle}| \pmod{p}$. Since p divides $|X|$, it must be that p divides $|X_{\langle \sigma \rangle}|$ also. Let us examine $X_{\langle \sigma \rangle}$. Now $(g_1, g_2, \cdots, g_p)$ is left fixed by σ, and hence by $\langle \sigma \rangle$, if and only if $g_1 = g_2 = \cdots = g_p$. We know at least one element in $X_{\langle \sigma \rangle}$, namely $(e, e, \cdots, e)$. Since p divides $|X_{\langle \sigma \rangle}|$, there must be at least p elements in $X_{\langle \sigma \rangle}$. Hence there exists some element $a \in G$, $a \neq e$, such that $(a, a, \cdots, a) \in X_{\langle \sigma \rangle}$ and hence $a^p = e$, so a has order p. Of course, $\langle a \rangle$ is a subgroup of G of order p. ◆

36.4 Corollary Let G be a finite group. Then G is a p-group if and only if $|G|$ is a power of p.

Proof We leave the proof of this corollary to Exercise 14. ◆

The Sylow Theorems

Let G be a group, and let $\mathscr{S}$ be the collection of all subgroups of G. We make $\mathscr{S}$ into a G-set by letting G act on $\mathscr{S}$ by conjugation. That is, if $H \in \mathscr{S}$ so $H \leq G$ and $g \in G$, then g acting on H yields the conjugate subgroup gHg^{-1}. (To avoid confusion, we will never write this action as gH.) Now $G_H = \{g \in G \mid gHg^{-1} = H\}$ is easily seen to be a subgroup of G (Exercise 11), and H is a normal subgroup of G_H. Since G_H consists of *all* elements of G that leave H invariant under conjugation, G_H is the largest subgroup of G having H as a normal subgroup.

36.5 Definition The subgroup G_H just discussed is the **normalizer of H in G** and will be denoted $N[H]$ from now on. ■

In the proof of the lemma that follows, we will use the fact that if H is a *finite* subgroup of a group G, then $g \in N[H]$ if $ghg^{-1} \in H$ for all $h \in H$. To see this, note that if $gh_1g^{-1} = gh_2g^{-1}$, then $h_1 = h_2$ by cancellation in the group G. Thus the conjugation map $i_g : H \to H$ given by $i_g(h) = ghg^{-1}$ is one to one. Because $|H|$ is finite, i_g must then map H *onto* H, so $gHg^{-1} = H$ and $g \in N[H]$.

36.6 Lemma Let H be a p-subgroup of a finite group G. Then

$$(N[H] : H) \equiv (G : H) \pmod{p}.$$

■ Historical Note

The Sylow theorems are due to the Norwegian mathematician Peter Ludvig Mejdell Sylow (1832–1918), who published them in a brief paper in 1872. Sylow stated the theorems in terms of permutation groups (since the abstract definition of a group had not yet been given). Georg Frobenius re-proved the theorems for abstract groups in 1887, even though he noted that in fact every group can be considered as a permutation group (Cayley's theorem [Theorem 8.16]). Sylow himself immediately applied the theorems to the question of solving algebraic equations and showed that any equation whose Galois group has order a power of a prime p is solvable by radicals.

Sylow spent most of his professional life as a high school teacher in Halden, Norway, and was only appointed to a position at Christiana University in 1898. He devoted eight years of his life to the project of editing the mathematical works of his countryman Niels Henrik Abel.

Proof Let $\mathscr{L}$ be the set of left cosets of H in G, and let H act on $\mathscr{L}$ by left translation, so that $h(xH) = (hx)H$. Then $\mathscr{L}$ becomes an H-set. Note that $|\mathscr{L}| = (G : H)$.

Let us determine $\mathscr{L}_H$, that is, those left cosets that are fixed under action by all elements of H. Now $xH = h(xH)$ if and only if $H = x^{-1}hxH$, or if and only if $x^{-1}hx \in H$. Thus $xH = h(xH)$ for all $h \in H$ if and only if $x^{-1}hx = x^{-1}h(x^{-1})^{-1} \in H$ for all $h \in H$, or if and only if $x^{-1} \in N[H]$ (see the comment before the lemma), or if and only if $x \in N[H]$. Thus the left cosets in $\mathscr{L}_H$ are those contained in $N[H]$. The number of such cosets is $(N[H] : H)$, so $|\mathscr{L}_H| = (N[H] : H)$.

Since H is a p-group, it has order a power of p by Corollary 36.4. Theorem 36.1 then tells us that $|\mathscr{L}| \equiv |\mathscr{L}_H| \pmod{p}$, that is, that $(G : H) \equiv (N[H] : H) \pmod{p}$. ◆

36.7 Corollary Let H be a p-subgroup of a finite group G. If p divides $(G : H)$, then $N[H] \neq H$.

Proof It follows from Lemma 36.6 that p divides $(N[H] : H)$, which must then be different from 1. Thus $H \neq N[H]$. ◆

We are now ready for the first of the Sylow theorems, which asserts the existence of prime-power subgroups of G for any prime power dividing $|G|$.

36.8 Theorem **(First Sylow Theorem)** Let G be a finite group and let $|G| = p^n m$ where $n \geq 1$ and where p does not divide m. Then

1. G contains a subgroup of order p^i for each i where $1 \leq i \leq n$,
2. Every subgroup H of G of order p^i is a normal subgroup of a subgroup of order p^{i+1} for $1 \leq i < n$.

Proof

1. We know G contains a subgroup of order p by Cauchy's theorem (Theorem 36.3). We use an induction argument and show that the existence of a subgroup of order p^i for $i < n$ implies the existence of a subgroup of order p^{i+1}. Let H be a subgroup of order p^i. Since $i < n$, we see p divides $(G : H)$. By Lemma 36.6, we then know p divides $(N[H] : H)$. Since H is a normal

subgroup of $N[H]$, we can form $N[H]/H$, and we see that p divides $|N[H]/H|$. By Cauchy's theorem, the factor group $N[H]/H$ has a subgroup K which is of order p. If $\gamma : N[H] \to N[H]/H$ is the canonical homomorphism, then $\gamma^{-1}[K] = \{x \in N[H] \mid \gamma(x) \in K\}$ is a subgroup of $N[H]$ and hence of G. This subgroup contains H and is of order p^{i+1}.

2. We repeat the construction in part 1 and note that $H < \gamma^{-1}[K] \leq N[H]$ where $|\gamma^{-1}[K]| = p^{i+1}$. Since H is normal in $N[H]$, it is of course normal in the possibly smaller group $\gamma^{-1}[K]$. ◆

36.9 Definition A **Sylow p-subgroup** P of a group G is a maximal p-subgroup of G, that is, a p-subgroup contained in no larger p-subgroup. ■

Let G be a finite group, where $|G| = p^n m$ as in Theorem 36.8. The theorem shows that the Sylow p-subgroups of G are precisely those subgroups of order p^n. If P is a Sylow p-subgroup, every conjugate gPg^{-1} of P is also a Sylow p-subgroup. The second Sylow theorem states that every Sylow p-subgroup can be obtained from P in this fashion; that is, any two Sylow p-subgroups are conjugate.

36.10 Theorem **(Second Sylow Theorem)** Let P_1 and P_2 be Sylow p-subgroups of a finite group G. Then P_1 and P_2 are conjugate subgroups of G.

Proof Here we will let one of the subgroups act on left cosets of the other, and use Theorem 36.1. Let $\mathscr{L}$ be the collection of left cosets of P_1, and let P_2 act on $\mathscr{L}$ by $y(xP_1) = (yx)P_1$ for $y \in P_2$. Then $\mathscr{L}$ is a P_2-set. By Theorem 36.1, $|\mathscr{L}_{P_2}| \equiv |\mathscr{L}| \pmod{p}$, and $|\mathscr{L}| = (G : P_1)$ is not divisible by p, so $|\mathscr{L}_{P_2}| \neq 0$. Let $xP_1 \in \mathscr{L}_{P_2}$. Then $yxP_1 = xP_1$ for all $y \in P_2$, so $x^{-1}yxP_1 = P_1$ for all $y \in P_2$. Thus $x^{-1}yx \in P_1$ for all $y \in P_2$, so $x^{-1}P_2x \leq P_1$. Since $|P_1| = |P_2|$, we must have $P_1 = x^{-1}P_2x$, so P_1 and P_2 are indeed conjugate subgroups. ◆

The final Sylow theorem gives information on the number of Sylow p-subgroups. A few illustrations are given after the theorem, and many more are given in the next section.

36.11 Theorem **(Third Sylow Theorem)** If G is a finite group and p divides $|G|$, then the number of Sylow p-subgroups is congruent to 1 modulo p and divides $|G|$.

Proof Let P be one Sylow p-subgroup of G. Let $\mathscr{S}$ be the set of all Sylow p-subgroups and let P act on $\mathscr{S}$ by conjugation, so that $x \in P$ carries $T \in \mathscr{S}$ into xTx^{-1}. By Theorem 36.1, $|\mathscr{S}| \equiv |\mathscr{S}_P| \pmod{p}$. Let us find $\mathscr{S}_P$. If $T \in \mathscr{S}_P$, then $xTx^{-1} = T$ for all $x \in P$. Thus $P \leq N[T]$. Of course $T \leq N[T]$ also. Since P and T are both Sylow p-subgroups of G, they are also Sylow p-subgroups of $N[T]$. But then they are conjugate in $N[T]$ by Theorem 36.10. Since T is a normal subgroup of $N[T]$, it is its only conjugate in $N[T]$. Thus $T = P$. Then $\mathscr{S}_P = \{P\}$. Since $|\mathscr{S}| \equiv |\mathscr{S}_P| = 1 \pmod{p}$, we see the number of Sylow p-subgroups is congruent to 1 modulo p.

Now let G act on $\mathscr{S}$ by conjugation. Since all Sylow p-subgroups are conjugate, there is only one orbit in $\mathscr{S}$ under G. If $P \in \mathscr{S}$, then $|\mathscr{S}| = |\text{orbit of } P| = (G : G_P)$ by Theorem 16.16. (G_P is, in fact, the normalizer of P.) But $(G : G_P)$ is a divisor of $|G|$, so the number of Sylow p-subgroups divides $|G|$. ◆

36.12 Example The Sylow 2-subgroups of S_3 have order 2. The subgroups of order 2 in S_3 in Example 8.7 are

$$\{\rho_0, \mu_1\}, \qquad \{\rho_0, \mu_2\}, \quad \{\rho_0, \mu_3\}.$$

Note that there are three subgroups and that $3 \equiv 1 \pmod 2$. Also, 3 divides 6, the order of S_3. We can readily check that

$$i_{\rho_2}[\{\rho_0, \mu_1\}] = \{\rho_0, \mu_3\} \qquad \text{and} \qquad i_{\rho_1}[\{\rho_0, \mu_1\}] = \{\rho_0, \mu_2\}$$

where $i_{\rho_j}(x) = \rho_j x \rho_j^{-1}$, illustrating that they are all conjugate. ▲

36.13 Example Let us use the Sylow theorems to show that no group of order 15 is simple. Let G have order 15. We claim that G has a normal subgroup of order 5. By Theorem 36.8 G has at least one subgroup of order 5, and by Theorem 36.11 the number of such subgroups is congruent to 1 modulo 5 and divides 15. Since 1, 6, and 11 are the only positive numbers less than 15 that are congruent to 1 modulo 5, and since among these only the number 1 divides 15, we see that G has exactly one subgroup P of order 5. But for each $g \in G$, the inner automorphism i_g of G with $i_g(x) = gxg^{-1}$ maps P onto a subgroup gPg^{-1}, again of order 5. Hence we must have $gPg^{-1} = P$ for all $g \in G$, so P is a normal subgroup of G. Therefore, G is not simple. (Example 37.10 will show that G must actually be abelian and therefore must be cyclic.) ▲

We trust that Example 36.13 gives some inkling of the power of Theorem 36.11. *Never underestimate a theorem that counts something, even modulo p.*

EXERCISES 36

Computations

In Exercises 1 through 4, fill in the blanks.

1. A Sylow 3-subgroup of a group of order 12 has order _____.
2. A Sylow 3-subgroup of a group of order 54 has order _____.
3. A group of order 24 must have either _____ or _____ Sylow 2-subgroups. (Use only the information given in Theorem 36.11.)
4. A group of order $255 = (3)(5)(17)$ must have either _____ or _____ Sylow 3-subgroups and _____ or _____ Sylow 5-subgroups. (Use only the information given in Theorem 36.11.)
5. Find all Sylow 3-subgroups of S_4 and demonstrate that they are all conjugate.
6. Find two Sylow 2-subgroups of S_4 and show that they are conjugate.

Concepts

In Exercises 7 through 9, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

7. Let p be a prime. A *p-group* is a group with the property that every element has order p.
8. The *normalizer* $N[H]$ of a subgroup H of a group G is the set of all inner automorphisms that carry H onto itself.

9. Let G be a group whose order is divisible by a prime p. The *Sylow p-subgroup* of a group is the largest subgroup P of G with the property that P has some power of p as its order.

10. Mark each of the following true or false.

_____**a.** Any two Sylow p-subgroups of a finite group are conjugate.

_____**b.** Theorem 36.11 shows that a group of order 15 has only one Sylow 5-subgroup.

_____**c.** Every Sylow p-subgroup of a finite group has order a power of p.

_____**d.** Every p-subgroup of every finite group is a Sylow p-subgroup.

_____**e.** Every finite abelian group has exactly one Sylow p-subgroup for each prime p dividing the order of G.

_____**f.** The normalizer in G of a subgroup H of G is always a normal subgroup of G.

_____**g.** If H is a subgroup of G, then H is always a normal subgroup of $N[H]$.

_____**h.** A Sylow p-subgroup of a finite group G is normal in G if and only if it is the only Sylow p-subgroup of G.

_____**i.** If G is an abelian group and H is a subgroup of G, then $N[H] = H$.

_____**j.** A group of prime-power order p^n has no Sylow p-subgroup.

Theory

11. Let H be a subgroup of a group G. Show that $G_H = \{g \in G \mid gHg^{-1} = H\}$ is a subgroup of G.

12. Let G be a finite group and let primes p and $q \neq p$ divide $|G|$. Prove that if G has precisely one proper Sylow p-subgroup, it is a normal subgroup, so G is not simple.

13. Show that every group of order 45 has a normal subgroup of order 9.

14. Prove Corollary 36.4.

15. Let G be a finite group and let p be a prime dividing $|G|$. Let P be a Sylow p-subgroup of G. Show that $N[N[P]] = N[P]$. [*Hint:* Argue that P is the only Sylow p-subgroup of $N[N[P]]$, and use Theorem 36.10.]

16. Let G be a finite group and let a prime p divide $|G|$. Let P be a Sylow p-subgroup of G and let H be any p-subgroup of G. Show there exists $g \in G$ such that $gHg^{-1} \leq P$.

17. Show that every group of order $(35)^3$ has a normal subgroup of order 125.

18. Show that there are no simple groups of order $255 = (3)(5)(17)$.

19. Show that there are no simple groups of order $p^r m$, where p is a prime, r is a positive integer, and $m < p$.

20. Let G be a finite group. Regard G as a G-set where G acts on itself by conjugation.

a. Show that G_G is the center $Z(G)$ of G. (See Section 15.)

b. Use Theorem 36.1 to show that the center of a finite nontrivial p-group is nontrivial.

21. Let p be a prime. Show that a finite group of order p^n contains *normal* subgroups H_i for $0 \leq i \leq n$ such that $|H_i| = p^i$ and $H_i < H_{i+1}$ for $0 \leq i < n$. [*Hint:* See Exercise 20 and get an idea from Section 35.]

22. Let G be a finite group and let P be a normal p-subgroup of G. Show that P is contained in every Sylow p-subgroup of G.

SECTION 37 APPLICATIONS OF THE SYLOW THEORY

In this section we give several applications of the Sylow theorems. It is intriguing to see how easily certain facts about groups of particular orders can be deduced. However, we should realize that we are working only with groups of finite order and really making

only a small dent in the general problem of determining the structure of all finite groups. If the order of a group has only a few factors, then the techniques illustrated in this section may be of some use in determining the structure of the group. This will be demonstrated further in Section 40, where we shall show how it is sometimes possible to describe all groups (up to isomorphism) of certain orders, even when some of the groups are not abelian. However, if the order of a finite group is highly composite, that is, has a large number of factors, the problem is in general much harder.

Applications to p-Groups and the Class Equation

37.1 Theorem Every group of prime-power order (that is, every finite p-group) is solvable.

Proof If G has order p^r, it is immediate from Theorem 36.8 that G has a subgroup H_i of order p^i normal in a subgroup H_{i+1} of order p^{i+1} for $1 \leq i < r$. Then

$$\{e\} = H_0 < H_1 < H_2 < \cdots < H_r = G$$

is a composition series, where the factor groups are of order p, and hence abelian and actually cyclic. Thus, G is solvable. ◆

The older proofs of the Sylow theorems used the *class equation*. The line of proof in Section 36 avoided explicit mention of the class equation, although Eq. (2) there is a general form of it. We now develop the classic class equation so you will be familiar with it.

Let X be a finite G-set where G is a finite group. Then Eq. (2) of Section 36 tells us that

$$|X| = |X_G| + \sum_{i=s+1}^{r} |Gx_i| \tag{1}$$

where x_i is an element in the ith orbit in X. Consider now the special case of Eq. (1), where $X = G$ and the action of G on G is by conjugation, so $g \in G$ carries $x \in X = G$ into gxg^{-1}. Then

$$\begin{aligned} X_G &= \{x \in G \mid gxg^{-1} = x \text{ for all } g \in G\} \\ &= \{x \in G \mid xg = gx \text{ for all } g \in G\} = Z(G), \end{aligned}$$

the center of G. If we let $c = |Z(G)|$ and $n_i = |Gx_i|$ in Eq. (1), then we obtain

$$|G| = c + n_{c+1} + \cdots + n_r \tag{2}$$

where n_i is the number of elements in the ith orbit of G under conjugation by itself. Note that n_i divides $|G|$ for $c + 1 \leq i \leq r$ since in Eq. (1) we know $|Gx_i| = (G : G_{x_i})$, which is a divisor of $|G|$.

37.2 Definition Equation (2) is the **class equation of** G. Each orbit in G under conjugation by G is a **conjugate class in** G. ■

37.3 Example It is readily checked that for S_3 of Example 8.7, the conjugate classes are

$$\{\rho_0\}, \qquad \{\rho_1, \rho_2\}, \qquad \{\mu_1, \mu_2, \mu_3\}.$$

The class equation of S_3 is

$$6 = 1 + 2 + 3. \qquad \blacktriangle$$

For illustration of the use of the class equation, we prove a theorem that Exercise 20(b) in Section 36 asked us to prove.

37.4 Theorem The center of a finite nontrivial p-group G is nontrivial.

Proof In Eq. (2) for G, each n_i divides $|G|$ for $c+1 \leq i \leq r$, so p divides each n_i, and p divides $|G|$. Therefore p divides c. Now $e \in Z(G)$, so $c \geq 1$. Therefore $c \geq p$, and there exists some $a \in Z(G)$ where $a \neq e$. ◆

We turn now to a lemma on direct products that will be used in some of the theorems that follow.

37.5 Lemma Let G be a group containing normal subgroups H and K such that $H \cap K = \{e\}$ and $H \vee K = G$. Then G is isomorphic to $H \times K$.

Proof We start by showing that $hk = kh$ for $k \in K$ and $h \in H$. Consider the commutator $hkh^{-1}k^{-1} = (hkh^{-1})k^{-1} = h(kh^{-1}k^{-1})$. Since H and K are normal subgroups of G, the two groupings with parentheses show that $hkh^{-1}k^{-1}$ is in both K and H. Since $K \cap H = \{e\}$, we see that $hkh^{-1}k^{-1} = e$, so $hk = kh$.

Let $\phi : H \times K \to G$ be defined by $\phi(h, k) = hk$. Then

$$\begin{aligned}\phi((h, k)(h', k')) &= \phi(hh', kk') = hh'kk' \\ &= hkh'k' = \phi(h, k)\phi(h', k'),\end{aligned}$$

so ϕ is a homomorphism.

If $\phi(h, k) = e$, then $hk = e$, so $h = k^{-1}$, and both h and k are in $H \cap K$. Thus $h = k = e$, so $\text{Ker}(\phi) = \{(e, e)\}$ and ϕ is one to one.

By Lemma 34.4, we know that $HK = H \vee K$, and $H \vee K = G$ by hypothesis. Thus ϕ is onto G, and $H \times K \simeq G$. ◆

37.6 Theorem For a prime number p, every group G of order p^2 is abelian.

Proof If G is not cyclic, then every element except e must be of order p. Let a be such an element. Then the cyclic subgroup $\langle a \rangle$ of order p does not exhaust G. Also let $b \in G$ with $b \notin \langle a \rangle$. Then $\langle a \rangle \cap \langle b \rangle = \{e\}$, since an element c in $\langle a \rangle \cap \langle b \rangle$ with $c \neq e$ would generate both $\langle a \rangle$ and $\langle b \rangle$, giving $\langle a \rangle = \langle b \rangle$, contrary to construction. From Theorem 36.8, $\langle a \rangle$ is normal in some subgroup of order p^2 of G, that is, normal in all of G. Likewise $\langle b \rangle$ is normal in G. Now $\langle a \rangle \vee \langle b \rangle$ is a subgroup of G properly containing $\langle a \rangle$ and of order dividing p^2. Hence $\langle a \rangle \vee \langle b \rangle$ must be all of G. Thus the hypotheses of Lemma 37.5 are satisfied, and G is isomorphic to $\langle a \rangle \times \langle b \rangle$ and therefore abelian. ◆

Further Applications

We turn now to a discussion of whether there exist simple groups of certain orders. We have seen that every group of prime order is simple. We also asserted that A_n is simple

for $n \geq 5$ and that A_5 is the smallest simple group that is not of prime order. There was a famous conjecture of Burnside that every finite simple group of nonprime order must be of even order. It was a triumph when this was proved by Thompson and Feit [21].

37.7 Theorem If p and q are distinct primes with $p < q$, then every group G of order pq has a single subgroup of order q and this subgroup is normal in G. Hence G is not simple. If q is not congruent to 1 modulo p, then G is abelian and cyclic.

Proof Theorems 36.8 and 36.11 tell us that G has a Sylow q-subgroup and that the number of such subgroups is congruent to 1 modulo q and divides pq, and therefore must divide p. Since $p < q$, the only possibility is the number 1. Thus there is only one Sylow q-subgroup Q of G. This group Q must be normal in G, for under an inner automorphism it would be carried into a group of the same order, hence itself. Thus G is not simple.

Likewise, there is a Sylow p-subgroup P of G, and the number of these divides pq and is congruent to 1 modulo p. This number must be either 1 or q. If q is not congruent to 1 modulo p, then the number must be 1 and P is normal in G. Let us assume that $q \not\equiv 1 \pmod{p}$. Since every element in Q other than e is of order q and every element in P other than e is of order p, we have $Q \cap P = \{e\}$. Also $Q \vee P$ must be a subgroup of G properly containing Q and of order dividing pq. Hence $Q \vee P = G$ and by Lemma 37.5 is isomorphic to $Q \times P$ or $\mathbb{Z}_q \times \mathbb{Z}_p$. Thus G is abelian and cyclic. ◆

We need another lemma for some of the counting arguments that follow.

37.8 Lemma If H and K are finite subgroups of a group G, then

$$|HK| = \frac{(|H|)(|K|)}{|H \cap K|}.$$

Proof Recall that $HK = \{hk \mid h \in H, k \in K\}$. Let $|H| = r$, $|K| = s$, and $|H \cap K| = t$. Now HK has at most rs elements. However, it is possible for h_1k_1 to equal h_2k_2, for $h_1, h_2 \in H$ and $k_1, k_2 \in K$; that is, there may be some collapsing. If $h_1k_1 = h_2k_2$, then let

$$x = (h_2)^{-1}h_1 = k_2(k_1)^{-1}$$

Now $x = (h_2)^{-1}h_1$ shows that $x \in H$, and $x = k_2(k_1)^{-1}$ shows that $x \in K$. Hence $x \in (H \cap K)$, and

$$h_2 = h_1x^{-1} \qquad \text{and} \qquad k_2 = xk_1.$$

On the other hand, if for $y \in (H \cap K)$ we let $h_3 = h_1y^{-1}$ and $k_3 = yk_1$, then clearly $h_3k_3 = h_1k_1$, with $h_3 \in H$ and $k_3 \in K$. Thus each element $hk \in HK$ can be represented in the form h_ik_i, for $h_i \in H$ and $k_i \in K$, as many times as there are elements of $H \cap K$, that is, t times. Therefore, the number of elements in HK is rs/t. ◆

Lemma 37.8 is another result that counts something, so do not underestimate it. The lemma will be used in the following way: A finite group G cannot have subgroups H and K that are too large with intersections that are too small, or the order of HK would have to exceed the order of G, which is impossible. For example, a group of order 24 cannot have two subgroups of orders 12 and 8 with an intersection of order 2.

The remainder of this section consists of several examples illustrating techniques of proving that all groups of certain orders are abelian or that they have nontrivial proper normal subgroups, that is, that they are not simple. We will use one fact we mentioned before only in the exercises. *A subgroup H of index 2 in a finite group G is always normal*, for by counting, we see that there are only the left cosets H itself and the coset consisting of all elements in G not in H. The right cosets are the same. Thus every right coset is a left coset, and H is normal in G.

37.9 Example No group of order p^r for $r > 1$ is simple, where p is a prime. For by Theorem 36.8 such a group G contains a subgroup of order p^{r-1} normal in a subgroup of order p^r, which must be all of G. Thus a group of order 16 is not simple; it has a normal subgroup of order 8. ▲

37.10 Example Every group of order 15 is cyclic (hence abelian and not simple, since 15 is not a prime). This is because $15 = (5)(3)$, and 5 is not congruent to 1 modulo 3. By Theorem 37.7 we are done. ▲

37.11 Example No group of order 20 is simple, for such a group G contains Sylow 5-subgroups in number congruent to 1 modulo 5 and a divisor of 20, hence only 1. This Sylow 5-subgroup is then normal, since all conjugates of it must be itself. ▲

37.12 Example No group of order 30 is simple. We have seen that if there is only one Sylow p-subgroup for some prime p dividing 30, we are done. By Theorem 36.11 the possibilities for the number of Sylow 5-subgroups are 1 or 6, and those for Sylow 3-subgroups are 1 or 10. But if G has six Sylow 5-subgroups, then the intersection of any two is a subgroup of each of order dividing 5, and hence just $\{e\}$. Thus each contains 4 elements of order 5 that are in none of the others. Hence G must contain 24 elements of order 5. Similarly, if G has 10 Sylow 3-subgroups, it has at least 20 elements of order 3. The two types of Sylow subgroups together would require at least 44 elements in G. Thus there is a normal subgroup either of order 5 or of order 3. ▲

37.13 Example No group of order 48 is simple. Indeed, we shall show that a group G of order 48 has a normal subgroup of either order 16 or order 8. By Theorem 36.11 G has either one or three Sylow 2-subgroups of order 16. If there is only one subgroup of order 16, it is normal in G, by a now familiar argument.

Suppose that there are three subgroups of order 16, and let H and K be two of them. Then $H \cap K$ must be of order 8, for if $H \cap K$ were of order ≤ 4, then by Lemma 37.8 HK would have at least $(16)(16)/4 = 64$ elements, contradicting the fact that G has only 48 elements. Therefore, $H \cap K$ is normal in both H and K (being of index 2, or by Theorem 36.8). Hence the normalizer of $H \cap K$ contains both H and K and must have order a multiple >1 of 16 and a divisor of 48, therefore 48. Thus $H \cap K$ must be normal in G. ▲

37.14 Example No group of order 36 is simple. Such a group G has either one or four subgroups of order 9. If there is only one such subgroup, it is normal in G. If there are four such subgroups, let H and K be two of them. As in Example 37.13, $H \cap K$ must have at least 3 elements, or HK would have to have 81 elements, which is impossible. Thus the normalizer of $H \cap K$ has as order a multiple of >1 of 9 and a divisor of 36; hence the order must

be either 18 or 36. If the order is 18, the normalizer is then of index 2 and therefore is normal in G. If the order is 36, then $H \cap K$ is normal in G. ▲

37.15 Example Every group of order $255 = (3)(5)(17)$ is abelian (hence cyclic by the Fundamental Theorem 11.12 and not simple, since 255 is not a prime). By Theorem 36.11 such a group G has only one subgroup H of order 17. Then G/H has order 15 and is abelian by Example 37.10. By Theorem 15.20, we see that the commutator subgroup C of G is contained in H. Thus as a subgroup of H, C has either order 1 or 17. Theorem 36.11 also shows that G has either 1 or 85 subgroups of order 3 and either 1 or 51 subgroups of order 5. However, 85 subgroups of order 3 would require 170 elements of order 3, and 51 subgroups of order 5 would require 204 elements of order 5 in G; both together would then require 375 elements in G, which is impossible. Hence there is a subgroup K having either order 3 or order 5 and normal in G. Then G/K has either order (5)(17) or order (3)(17), and in either case Theorem 37.7 shows that G/K is abelian. Thus $C \leq K$ and has order either 3, 5, or 1. Since $C \leq H$ showed that C has order 17 or 1, we conclude that C has order 1. Hence $C = \{e\}$, and $G/C \simeq G$ is abelian. The Fundamental Theorem 11.12 then shows that G is cyclic. ▲

EXERCISES 37

Computations

1. Let D_4 be the group of symmetries of the square in Example 8.10.
   a. Find the decomposition of D_4 into conjugate classes.
   b. Write the class equation for D_4.
2. By arguments similar to those used in the examples of this section, convince yourself that every nontrivial group of order not a prime and less than 60 contains a nontrivial proper normal subgroup and hence is not simple. You need not write out the details. (The hardest cases were discussed in the examples.)

Concepts

3. Mark each of the following true or false.
   _____ **a.** Every group of order 159 is cyclic.
   _____ **b.** Every group of order 102 has a nontrivial proper normal subgroup.
   _____ **c.** Every solvable group is of prime-power order.
   _____ **d.** Every group of prime-power order is solvable.
   _____ **e.** It would become quite tedious to show that no group of nonprime order between 60 and 168 is simple by the methods illustrated in the text.
   _____ **f.** No group of order 21 is simple.
   _____ **g.** Every group of 125 elements has at least 5 elements that commute with every element in the group.
   _____ **h.** Every group of order 42 has a normal subgroup of order 7.
   _____ **i.** Every group of order 42 has a normal subgroup of order 8.
   _____ **j.** The only simple groups are the groups $\mathbb{Z}_p$ and A_n where p is a prime and $n \neq 4$.

Theory

4. Prove that every group of order (5)(7)(47) is abelian and cyclic.
5. Prove that no group of order 96 is simple.
6. Prove that no group of order 160 is simple.
7. Show that every group of order 30 contains a subgroup of order 15. [*Hint:* Use the last sentence in Example 37.12, and go to the factor group.]
8. This exercise determines the conjugate classes of S_n for every integer $n \geq 1$.
   a. Show that if $\sigma = (a_1, a_2, \cdots, a_m)$ is a cycle in S_n and τ is any element of S_n then $\tau\sigma\tau^{-1} = (\tau a_1, \tau a_2, \cdots, \tau a_m)$.
   b. Argue from (a) that any two cycles in S_n of the same length are conjugate.
   c. Argue from (a) and (b) that a product of s disjoint cycles in S_n of lengths r_i for $i = 1, 2, \cdots, s$ is conjugate to every other product of s disjoint cycles of lengths r_i in S_n.
   d. Show that the number of conjugate classes in S_n is $p(n)$, where $p(n)$ is the number of ways, neglecting the order of the summands, that n can be expressed as a sum of positive integers. The number $p(n)$ is the **number of partitions of** n.
   e. Compute $p(n)$ for $n = 1, 2, 3, 4, 5, 6, 7$.
9. Find the conjugate classes and the class equation for S_4. [*Hint:* Use Exercise 8.]
10. Find the class equation for S_5 and S_6. [*Hint:* Use Exercise 8.]
11. Show that the number of conjugate classes in S_n is also the number of different abelian groups (up to isomorphism) of order p^n, where p is a prime number. [*Hint:* Use Exercise 8.]
12. Show that if $n > 2$, the center of S_n is the subgroup consisting of the identity permutation only. [*Hint:* Use Exercise 8.]

SECTION 38 FREE ABELIAN GROUPS

In this section we introduce the concept of free abelian groups and prove some results concerning them. The section concludes with a demonstration of the Fundamental Theorem of finitely generated abelian groups (Theorem 11.12).

Free Abelian Groups

We should review the notions of a generating set for a group G and a finitely generated group, as given in Section 7. In this section we shall deal exclusively with abelian groups and use additive notations as follows:

$$0 \text{ for the identity, } + \text{ for the operation,}$$

$$\left.\begin{aligned} na &= \underbrace{a + a + \cdots + a}_{n \text{ summands}} \\ -na &= \underbrace{(-a) + (-a) + \cdots + (-a)}_{n \text{ summands}} \end{aligned}\right\} \text{ for } n \in \mathbb{Z}^+ \text{ and } a \in G.$$

$$0a = 0 \text{ for the first } 0 \text{ in } \mathbb{Z} \text{ and the second in } G.$$

We shall continue to use the symbol $\times$ for direct product of groups rather than change to direct sum notation.

Notice that $\{(1, 0), (0, 1)\}$ is a generating set for the group $\mathbb{Z} \times \mathbb{Z}$ since $(n, m) = n(1, 0) + m(0, 1)$ for any (n, m) in $\mathbb{Z} \times \mathbb{Z}$. This generating set has the property that each element of $\mathbb{Z} \times \mathbb{Z}$ can be *uniquely* expressed in the form $n(1, 0) + m(0, 1)$. That is, the coefficients n and m in $\mathbb{Z}$ are unique.

38.1 Theorem Let X be a subset of a nonzero abelian group G. The following conditions on X are equivalent.

1. Each nonzero element a in G can be expressed *uniquely* (up to order of summands) in the form $a = n_1x_1 + n_2x_2 + \cdots + n_rx_r$ for $n_i \neq 0$ in $\mathbb{Z}$ and distinct x_i in X.
2. X generates G, and $n_1x_1 + n_2x_2 + \cdots + n_rx_r = 0$ for $n_i \in \mathbb{Z}$ and distinct $x_i \in X$ if and only if $n_1 = n_2 = \cdots = n_r = 0$.

Proof Suppose Condition 1 is true. Since $G \neq \{0\}$, we have $X \neq \{0\}$. It follows from 1 that $0 \notin X$, for if $x_i = 0$ and $x_j \neq 0$, then $x_j = x_i + x_j$, which would contradict the uniqueness of the expression for x_j. From Condition 1, X generates G, and $n_1x_1 + n_2x_2 + \cdots + n_rx_r = 0$ if $n_1 = n_2 = \cdots = n_r = 0$. Suppose that $n_1x_1 + n_2x_2 + \cdots + n_rx_r = 0$ with some $n_i \neq 0$; by dropping terms with zero coefficients and renumbering, we can assume all $n_i \neq 0$. Then

$$\begin{aligned} x_1 &= x_1 + (n_1x_1 + n_2x_2 + \cdots + n_rx_r) \\ &= (n_1 + 1)x_1 + n_2x_2 + \cdots + n_rx_r, \end{aligned}$$

which gives two ways of writing $x_1 \neq 0$, contradicting the uniqueness assumption in Condition 1. Thus Condition 1 implies Condition 2.

We now show that Condition 2 implies Condition 1. Let $a \in G$. Since X generates G, we see a can be written in the form $a = n_1x_1 + n_2x_2 + \cdots + n_rx_r$. Suppose a has another such expression in terms of elements of X. By using some zero coefficients in the two expressions, we can assume they involve the same elements in X and are of the form

$$\begin{aligned} a &= n_1x_1 + n_2x_2 + \cdots n_rx_r \\ a &= m_1x_1 + m_2x_2 + \cdots m_rx_r. \end{aligned}$$

Subtracting, we obtain

$$0 = (n_1 - m_1)x_1 + (n_2 - m_2)x_2 + \cdots + (n_r - m_r)x_r,$$

so $n_i - m_i = 0$ by Condition 2, and $n_i = m_i$ for $i = 1, 2, \cdots, r$. Thus the coefficients are unique. ◆

38.2 Definition An abelian group having a generating set X satisfying the conditions described in Theorem 38.1 is a **free abelian group**, and X is a **basis** for the group. ■

38.3 Example The group $\mathbb{Z} \times \mathbb{Z}$ is free abelian and $\{(1, 0), (0, 1)\}$ is a basis. Similarly, a basis for the free abelian group $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$, and so on. Thus finite direct products of the group $\mathbb{Z}$ with itself are free abelian groups. ▲

38.4 Example The group $\mathbb{Z}_n$ is not free abelian, for $nx = 0$ for every $x \in \mathbb{Z}_n$, and $n \neq 0$, which would contradict Condition 2. ▲

Suppose a free abelian group G has a finite basis $X = \{x_1, x_2, \cdots, x_r\}$. If $a \in G$ and $a \neq 0$, then a has a *unique* expression of the form

$$a = n_1x_1 + n_2x_2 + \cdots + n_rx_r \quad \text{for} \quad n_i \in \mathbb{Z}.$$

(Note that in the preceding expression for a, we included all elements x_i of our finite basis X, as opposed to the expression for a in Condition 1 of Theorem 38.1 where the basis may be infinite. Thus in the preceding expression for a we must allow the possibility that some of the coefficients n_i are zero, whereas in Condition 1 of Theorem 38.1, we specified that each $n_i \neq 0$.)
We define

$$\phi : G \to \underbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}_{r \text{ factors}}$$

by $\phi(a) = (n_1, n_2, \cdots, n_r)$ and $\phi(0) = (0, 0, \cdots, 0)$. It is straightforward to check that ϕ is an isomorphism. We leave the details to the exercises (see Exercise 9) and state the result as a theorem.

38.5 Theorem If G is a nonzero free abelian group with a basis of r elements, then G is isomorphic to $\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$ for r factors.

It is a fact that any two bases of a free abelian group G contain the same number of elements. We shall prove this only if G has a finite basis, although it is also true if every basis of G is infinite. The proof is really lovely; it gives an easy characterization of the number of elements in a basis in terms of the size of a factor group.

38.6 Theorem Let $G \neq \{0\}$ be a free abelian group with a finite basis. Then every basis of G is finite, and all bases of G have the same number of elements.

Proof Let G have a basis $\{x_1, x_2, \cdots, x_r\}$. Then G is isomorphic to $\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$ for r factors. Let $2G = \{2g \mid g \in G\}$. It is readily checked that $2G$ is a subgroup of G. Since $G \simeq \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$ for r factors, we have

$$\begin{aligned} G/2G &\simeq (\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z})/(2\mathbb{Z} \times 2\mathbb{Z} \times \cdots \times 2\mathbb{Z}) \\ &\simeq \mathbb{Z}_2 \times \mathbb{Z}_2 \times \cdots \times \mathbb{Z}_2 \end{aligned}$$

for r factors. Thus $|G/2G| = 2^r$, so the number of elements in any finite basis X is $\log_2 |G/2G|$. Thus any two finite bases have the same number of elements.

It remains to show that G cannot also have an infinite basis. Let Y be any basis for G, and let $\{y_1, y_2, \cdots, y_s\}$ be distinct elements in Y. Let H be the subgroup of G generated by $\{y_1, y_2, \cdots, y_s\}$, and let K be the subgroup of G generated by the remaining elements of Y. It is readily checked that $G \simeq H \times K$, so

$$G/2G \simeq (H \times K)/(2H \times 2K) \simeq (H/2H) \times (K/2K).$$

Since $|H/2H| = 2^s$, we see $|G/2G| \geq 2^s$. Since we have $|G/2G| = 2^r$, we see that $s \leq r$. Then Y cannot be an infinite set, for we could take $s > r$. ◆

38.7 Definition If G is a free abelian group, the **rank** of G is the number of elements in a basis for G. (All bases have the same number of elements.) ■

Proof of the Fundamental Theorem

We shall prove the Fundamental Theorem (Theorem 11.12) by showing that any finitely generated abelian group is isomorphic to a factor group of the form

$$(\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z})/(d_1\mathbb{Z} \times d_2\mathbb{Z} \times \cdots \times d_s\mathbb{Z} \times \{0\} \times \cdots \times \{0\}),$$

where both "numerator" and "denominator" have n factors, and d_1 divides d_2, which divides $d_3 \cdots$, which divides d_s. The prime-power decomposition of Theorem 11.12 will then follow.

To show that G is isomorphic to such a factor group, we will show that there is a homomorphism of $\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$ onto G with kernel of the form $d_1\mathbb{Z} \times d_2\mathbb{Z} \times \cdots \times d_s\mathbb{Z} \times \{0\} \times \cdots \times \{0\}$. The result will then follow by Theorem 14.11. The theorems that follow give the details of the argument. Our purpose in these introductory paragraphs is to let us see where we are going as we read what follows.

38.8 Theorem Let G be a finitely generated abelian group with generating set $\{a_1, a_2, \cdots, a_n\}$. Let

$$\phi : \underbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}_{n \text{ factors}} \to G$$

be defined by $\phi(h_1, h_2, \cdots, h_n) = h_1a_1 + h_2a_2 + \cdots + h_na_n$. Then ϕ is a homomorphism onto G.

Proof From the meaning of h_ia_i for $h_i \in \mathbb{Z}$ and $a_i \in G$, we see at once that

$$\begin{aligned}
\phi[(h_1, \cdots, h_n) + (k_1, \cdots, k_n)] &= \phi(h_1 + k_1, \cdots, h_n + k_n) \\
&= (h_1 + k_1)a_1 + \cdots + (h_n + k_n)a_n \\
&= (h_1a_1 + k_1a_1) + \cdots + (h_na_n + k_na_n) \\
&= (h_1a_1 + \cdots + h_na_n) + (k_1a_1 + \cdots + k_na_n) \\
&= \phi(k_1, \cdots, k_n) + \phi(h_1, \cdots, h_n).
\end{aligned}$$

Since $\{a_1, \cdots, a_n\}$ generates G, clearly the homomorphism ϕ is onto G. ◆

We now prove a "replacement property" that makes it possible for us to adjust a basis.

38.9 Theorem If $X = \{x_1, \cdots, x_r\}$ is a basis for a free abelian group G and $t \in \mathbb{Z}$, then for $i \neq j$, the set

$$Y = \{x_1, \cdots, x_{j-1}, x_j + tx_i, x_j, x_{j+1}, \cdots, x_r\}$$

is also a basis for G.

Proof Since $x_j = (-t)x_i + (1)(x_j + tx_i)$, we see that x_j can be recovered from Y, which thus also generates G. Suppose

$$n_1x_1 + \cdots + n_{j-1}x_{j-1} + n_j(x_j + tx_i) + n_{j+1}x_{j+1} + \cdots + n_rx_r = 0.$$

Then

$$n_1x_1 + \cdots + (n_i + n_jt)x_i + \cdots + n_jx_j + \cdots + n_rx_r = 0.$$

and since X is a basis, $n_1 = \cdots = n_i + n_jt = \cdots = n_j = \cdots = n_r = 0$. From $n_j = 0$ and $n_i + n_jt = 0$, it follows that $n_i = 0$ also, so $n_1 = \cdots = n_i = \cdots = n_j = \cdots = n_r = 0$, and Condition 2 of Theorem 38.1 is satisfied. Thus Y is a basis. ◆

38.10 Example A basis for $\mathbb{Z} \times \mathbb{Z}$ is $\{(1, 0), (0, 1)\}$. Another basis is $\{(1, 0), (4, 1)\}$ for $(4, 1) = 4(1, 0) + (0, 1)$. However, $\{(3, 0), (0, 1)\}$ is not a basis. For example, we cannot express $(2, 0)$ in the form $n_1(3, 0) + n_2(0, 1)$, for $n, n_2 \in \mathbb{Z}$. Here $(3, 0) = (1, 0) + 2(1, 0)$, and a multiple of a basis element was added to *itself*, rather than to a *different* basis element. ▲

A free abelian group G of finite rank may have many bases. We show that if $K \leq G$, then K is also free abelian with rank not exceeding that of G. Equally important, there exist bases of G and K nicely related to each other.

38.11 Theorem Let G be a nonzero free abelian group of finite rank n, and let K be a nonzero subgroup of G. Then K is free abelian of rank $s \leq n$. Furthermore, there exists a basis $\{x_1, x_2, \cdots, x_n\}$ for G and positive integers, $d_1, d_2, \cdots, d_s$ where d_i divides d_{i+1} for $i = 1, \cdots, s - 1$, such that $\{d_1x_1, d_2x_2, \cdots, d_sx_s\}$ is a basis for K.

Proof We show that K has a basis of the described form, which will show that K is free abelian of rank at most n. Suppose $Y = \{y_1, \cdots, y_n\}$ is a basis for G. All nonzero elements in K can be expressed in the form

$$k_1y_1 + \cdots + k_ny_n,$$

where some $|k_i|$ is nonzero. Among *all* bases Y for G, select one Y_1 that yields the minimal such nonzero value $|k_i|$ as all nonzero elements of K are written in terms of the basis elements in Y_1. By renumbering the elements of Y_1 if necessary, we can assume there is $w_1 \in K$ such that

$$w_1 = d_1y_1 + k_2y_2 + \cdots + k_ny_n$$

where $d_1 > 0$ and d_1 is the minimal attainable coefficient as just described. Using the division algorithm, we write $k_j = d_1q_j + r_j$ where $0 \leq r_j < d_1$ for $j = 2, \cdots, n$. Then

$$w_1 = d_1(y_1 + q_2y_2 + \cdots + q_ny_n) + r_2y_2 + \cdots + r_ny_n. \tag{1}$$

Now let $x_1 = y_1 + q_2y_2 + \cdots + q_ny_n$. By Theorem 38.9 $\{x_1, y_2, \cdots, y_n\}$ is also a basis for G. From Eq. (1) and our choice of Y_1 for minimal coefficient d_1, we see that $r_2 = \cdots = r_n = 0$. Thus $d_1x_1 \in K$.

We now consider bases for G of the form $\{x_1, y_2, \cdots, y_n\}$. Each element of K can be expressed in the form

$$h_1x_1 + k_2y_2 + \cdots + k_ny_n.$$

Since $d_1x_1 \in K$, we can subtract a suitable multiple of d_1x_1 and then using the minimality of d_1 to see that h_1 is a multiple of d_1, we see we actually have $k_2y_2 + \cdots + k_ny_n$ in K. Among all such bases $\{x_1, y_2, \cdots, y_n\}$, we choose one Y_2 that leads to some $k_i \neq 0$ of minimal magnitude. (It is possible all k_i are always zero. In this case, K is generated by d_1x_1 and we are done.) By renumbering the elements of Y_2 we can assume that there is $w_2 \in K$ such that

$$w_2 = d_2y_2 + \cdots + k_ny_n$$

where $d_2 < 0$ and d_2 is minimal as just described. Exactly as in the preceding paragraph, we can modify our basis from $Y_2 = \{x_1, y_2, \cdots, y_n\}$ to a basis $\{x_1, x_2, y_3, \cdots, y_n\}$ for G where $d_1x_1 \in K$ and $d_2x_2 \in K$. Writing $d_2 = d_1q + r$ for $0 \leq r < d_1$, we see that $\{x_1 + qx_2, x_2, y_3, \cdots, y_n\}$ is a basis for G, and $d_1x_1 + d_2x_2 = d_1(x_1 + qx_2) + rx_2$ is in K. By our minimal choice of d_1, we see $r = 0$, so d_1 divides d_2.

We now consider all bases of the form $\{x_1, x_2, y_3, \cdots, y_n\}$ for G and examine elements of K of the form $k_3y_3 + \cdots + k_ny_n$. The pattern is clear. The process continues until we obtain a basis $\{x_1, x_2, \cdots, x_s, y_{s+1}, \cdots, y_n\}$ where the only element of K of the form $k_{s+1}y_{s+1} + \cdots + k_ny_n$ is zero, that is, all k_i are zero. We then let $x_{s+1} = y_{s+1}, \cdots, x_n = y_n$ and obtain a basis for G of the form described in the statement of Theorem 38.11. ◆

38.12 Theorem Every finitely generated abelian group is isomorphic to a group of the form

$$\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_r} \times \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z},$$

where m_i divides m_{i+1} for $i = 1, \cdots, r-1$.

Proof For the purposes of this proof, it will be convenient to use as notations $\mathbb{Z}/1\mathbb{Z} = \mathbb{Z}/\mathbb{Z} \simeq \mathbb{Z}_1 = \{0\}$. Let G be finitely generated by n elements. Let $F = \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$ for n factors. Consider the homomorphism $\phi : F \to G$ of Theorem 38.8, and let K be the kernel of this homomorphism. Then there is a basis for F of the form $\{x_1, \cdots, x_n\}$, where $\{d_1x_1, \cdots, d_sx_s\}$ is a basis for K and d_i divides d_{i+1} for $i = 1, \cdots, s-1$. By Theorem 14.11, G is isomorphic to F/K. But

$$\begin{aligned} F/K &\simeq (\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z})/(d_1\mathbb{Z} \times d_2\mathbb{Z} \times \cdots \times d_s\mathbb{Z} \times \{0\} \times \cdots \times \{0\}) \\ &\simeq \mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \cdots \times \mathbb{Z}_{d_s} \times \mathbb{Z} \times \cdots \times \mathbb{Z}. \end{aligned}$$

It is possible that $d_1 = 1$, in which case $\mathbb{Z}_{d_1} = \{0\}$ and can be dropped (up to isomorphism) from this product. Similarly, d_2 may be 1, and so on. We let m_1 be the first $d_i > 1$, m_2 be the next d_i, and so on, and our theorem follows at once. ◆

We have demonstrated the toughest part of the Fundamental Theorem (Theorem 11.12). Of course, a prime-power decomposition exists since we can break the groups $\mathbb{Z}_{m_i}$ into prime-power factors. The only remaining part of Theorem 11.12 concerns the

uniqueness of the Betti number, of the torsion coefficients, and of the prime powers. The Betti number appears as the rank of the free abelian group G/T, where T is the torsion subgroup of G. This rank is invariant by Theorem 38.6 which shows the uniqueness of the Betti number. The uniqueness of the torsion coefficients and of prime powers is a bit more difficult to show. We give some exercises that indicate their uniqueness (see Exercises 14 through 22).

EXERCISES 38

Computations

1. Find a basis $\{(a_1, a_2, a_3), (b_1, b_2, b_3), (c_1, c_2, c_3)\}$ for $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ with all $a_i \neq 0$, all $b_i \neq 0$, and all $c_i \neq 0$. (Many answers are possible.)
2. Is $\{(2, 1), (3, 1)\}$ a basis for $\mathbb{Z} \times \mathbb{Z}$? Prove your assertion.
3. Is $\{(2, 1), (4, 1)\}$ a basis for $\mathbb{Z} \times \mathbb{Z}$? Prove your assertion.
4. Find conditions on $a, b, c, d \in \mathbb{Z}$ for $\{(a, b), (c, d)\}$ to be a basis for $\mathbb{Z} \times \mathbb{Z}$. [*Hint:* Solve $x(a, b) + y(c, d) = (e, f)$ in $\mathbb{R}$, and see when the x and y lie in $\mathbb{Z}$.]

Concepts

In Exercises 5 and 6, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

5. The *rank* of a free abelian group G is the number of elements in a generating set for G.
6. A *basis* for a nonzero abelian group G is a generating set $X \subseteq G$ such that $n_1x_1 + n_2x_2 + \cdots + n_mx_m = 0$ for distinct $x_i \in X$ and $n_i \in \mathbb{Z}$ only if $n_1 = n_2 = \cdots = n_m = 0$.
7. Show by example that it is possible for a proper subgroup of a free abelian group of finite rank r also to have rank r.
8. Mark each of the following true or false.

   _____ **a.** Every free abelian group is torsion free.
   _____ **b.** Every finitely generated torsion-free abelian group is a free abelian group.
   _____ **c.** There exists a free abelian group of every positive integer rank.
   _____ **d.** A finitely generated abelian group is free abelian if its Betti number equals the number of elements in some generating set.
   _____ **e.** If X generates a free abelian group G and $X \subseteq Y \subseteq G$, then Y generates G.
   _____ **f.** If X is a basis for a free abelian group G and $X \subseteq Y \subseteq G$, then Y is a basis for G.
   _____ **g.** Every nonzero free abelian group has an infinite number of bases.
   _____ **h.** Every free abelian group of rank at least 2 has an infinite number of bases.
   _____ **i.** If K is a nonzero subgroup of a finitely generated free abelian group, then K is free abelian.
   _____ **j.** If K is a nonzero subgroup of a finitely generated free abelian group, then G/K is free abelian.

Theory

9. Complete the proof of Theorem 38.5 (See the two sentences preceding the theorem).
10. Show that a free abelian group contains no nonzero elements of finite order.

11. Show that if G and G' are free abelian groups, then $G \times G'$ is free abelian.

12. Show that free abelian groups of finite rank are precisely the finitely generated abelian groups containing no nonzero elements of finite order.

13. Show that $\mathbb{Q}$ under addition is not a free abelian group. [*Hint:* Show that no two distinct rational numbers n/m and r/s could be contained in a set satisfying Condition 2 of Thorem 38.1.]

Exercises 14 through 19 deal with showing the uniqueness of the prime powers appearing in the prime-power decomposition of the torsion subgroup T of a finitely generated abelian group.

14. Let p be a fixed prime. Show that the elements of T having as order some power of p, together with zero, form a subgroup T_p of T.

15. Show that in any prime-power decomposition of T, the subgroup T_p in the preceding exercise is isomorphic to the direct product of those cyclic factors of order some power of the prime p. [This reduces our problem to showing that the group T_p cannot have essentially different decompositions into products of cyclic groups.]

16. Let G be any abelian group and let n be any positive integer. Show that $G[n] = \{x \in G \mid nx = 0\}$ is a subgroup of G. (In multiplicative notation, $G[n] = \{x \in G \mid x^n = e\}$.)

17. Referring to Exercise 16, show that $\mathbb{Z}_{p^r}[p] \simeq \mathbb{Z}_p$ for any $r \geq 1$ and prime p.

18. Using Exercise 17, show that

$$(\mathbb{Z}_{p^{r_1}} \times \mathbb{Z}_{p^{r_2}} \times \cdots \times \mathbb{Z}_{p^{r_m}})[p] \simeq \underbrace{\mathbb{Z}_p \times \mathbb{Z}_p \times \cdots \times \mathbb{Z}_p}_{m \text{ factors}}$$

provided each $r_i \geq 1$.

19. Let G be a finitely generated abelian group and T_p the subgroup defined in Exercise 14. Suppose $T_p \simeq \mathbb{Z}_{p^{r_1}} \times \mathbb{Z}_{p^{r_2}} \times \cdots \times \mathbb{Z}_{p^{r_m}} \simeq \mathbb{Z}_{p^{s_1}} \mathbb{Z}_{p^{s_2}} \times \cdots \times \mathbb{Z}_{p^{s_n}}$, where $1 \leq r_1 \leq r_2 \leq \cdots \leq r_m$ and $1 \leq s_1 \leq s_2 \leq \cdots \leq s_n$. We need to show that $m = n$ and $r_i = s_i$ for $i = 1, \cdots, n$ to complete the demonstration of uniqueness of the prime-power decomposition.

a. Use Exercise 18 to show that $n = m$.

b. Show $r_1 = s_1$. Suppose $r_i = s_i$ for all $i < j$. Show $r_j = s_j$, which will complete the proof. [*Hint:* Suppose $r_j < s_j$. Consider the subgroup $p^{r_j} T_p = \{p^{r_j} x \mid x \in T_p\}$, and show that this subgroup would then have two prime-power decompositions involving different numbers of nonzero factors. Then argue that this is impossible by part (a) of this exercise.]

Let T be the torsion subgroup of a finitely generated abelian group. Suppose $T \simeq \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_r} \simeq \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_s}$, where m_i divides m_{i+1} for $i = 1, \cdots, r-1$, and n_j divides n_{j+1} for $n = 1, \cdots, s-1$, and $m_1 > 1$ and $n_1 > 1$. We wish to show that $r = s$ and $m_k = n_k$ for $k = 1, \cdots, r$, demonstrating the uniqueness of the torsion coefficients. This is done in Exercises 20 through 22.

20. Indicate how a prime-power decomposition can be obtained from a torsion-coefficient decomposition. (Observe that the preceding exercises show the prime powers obtained are unique.)

21. Argue from Exercise 20 that m_r and n_s can both be characterized as follows. Let $p_1, \cdots, p_t$ be the distinct primes dividing $|T|$, and let $p_1^{h_1}, \cdots, p_t^{h_t}$ be the highest powers of these primes appearing in the (unique) prime-power decomposition. Then $m_r = n_s = p_1^{h_1} p_2^{h_2} \cdots p_t^{h_t}$.

22. Characterize m_{r-1} and n_{s-1}, showing that they are equal, and continue to show $m_{r-i} = n_{s-1}$ for $i = 1, \cdots, r-1$, and then $r = s$.

SECTION 39 FREE GROUPS

In this section and Section 40 we discuss a portion of group theory that is of great interest not only in algebra but in topology as well. In fact, an excellent and readable discussion of free groups and presentations of groups is found in Crowell and Fox [46, Chapters 3 and 4].

Words and Reduced Words

Let A be any (not necessarily finite) set of elements a_i for $i \in I$. We think of A as an **alphabet** and of the a_i as **letters** in the alphabet. Any symbol of the form a_i^n with $n \in \mathbb{Z}$ is a **syllable** and a finite string w of syllables written in juxtaposition is a **word.** We also introduce the **empty word** 1, which has no syllables.

39.1 Example Let $A = \{a_1, a_2, a_3\}$. Then

$$a_1 a_3^{-4} a_2^2 a_3, \qquad a_2^3 a_2^{-1} a_3 a_1^2 a_1^{-7}, \qquad \text{and} \qquad a_3^2$$

are all words, if we follow the convention of understanding that a_i^1 is the same as a_i. ▲

There are two natural types of modifications of certain words, the **elementary contractions.** The first type consists of replacing an occurrence of $a_i^m a_i^n$ in a word by a_i^{m+n}. The second type consists of replacing an occurrence of a_i^0 in a word by 1, that is, dropping it out of the word. By means of a finite number of elementary contractions, every word can be changed to a **reduced word,** one for which no more elementary contractions are possible. Note that these elementary contractions formally amount to the usual manipulations of integer exponents.

39.2 Example The reduced form of the word $a_2^3 a_2^{-1} a_3 a_1^2 a_1^{-7}$ of Example 39.1 is $a_2^2 a_3 a_1^{-5}$. ▲

It should be said here once and for all that we are going to gloss over several points that some books spend pages proving, usually by complicated induction arguments broken down into many cases. For example, suppose we are given a word and wish to find its reduced form. There may be a variety of elementary contractions that could be performed first. How do we know that the reduced word we end up with is the same no matter in what order we perform the elementary contractions? The student will probably say this is obvious. Some authors spend considerable effort proving this. The author tends to agree here with the student. Proofs of this sort he regards as tedious, and they have never made him more comfortable about the situation. However, the author is the first to acknowledge that he is not a great mathematician. In deference to the fact that many mathematicians feel that these things do need considerable discussion, we shall mark an occasion when we just state such facts by the phrase, "It would seem obvious that," keeping the quotation marks.

Free Groups

Let the set of all reduced words formed from our alphabet A be $F[A]$. We now make $F[A]$ into a group in a natural way. For w_1 and w_2 in $F[A]$, define $w_1 \cdot w_2$ to be the reduced form of the word obtained by the juxtaposition $w_1 w_2$ of the two words.

39.3 Example If

$$w_1 = {a_2}^3 {a_1}^{-5} {a_3}^2$$

and

$$w_2 = {a_3}^{-2} {a_1}^2 a_3 {a_2}^{-2},$$

then $w_1 \cdot w_2 = {a_2}^3 {a_1}^{-3} a_3 {a_2}^{-2}$. ▲

"It would seem obvious that" this operation of multiplication on $F[A]$ is well defined and associative. The empty word 1 acts as an identity element. "It would seem obvious that" given a reduced word $w \in F[A]$, if we form the word obtained by first writing the syllables of w in the opposite order and second by replacing each ${a_i}^n$ by ${a_i}^{-n}$, then the resulting word w^{-1} is a reduced word also, and

$$w \cdot w^{-1} = w^{-1} \cdot w = 1.$$

39.4 Definition The group $F[A]$ just described is the **free group generated** by A. ■

Look back at Theorem 7.6 and the definition preceding it to see that the present use of the term *generated* is consistent with the earlier use.

Starting with a group G and a generating set $\{a_i \mid i \in I\}$ which we will abbreviate by $\{a_i\}$, we might ask if G is *free* on $\{a_i\}$, that is, if G is essentially the free group generated by $\{a_i\}$. We define precisely what this is to mean.

39.5 Definition If G is a group with a set $A = \{a_i\}$ of generators, and if G is isomorphic to $F[A]$ under a map $\phi : G \to F[A]$ such that $\phi(a_i) = a_i$, then G is **free on** A, and the a_i are **free generators of** G. A group is **free** if it is free on some nonempty set A. ■

39.6 Example The only example of a free group that has occurred before is $\mathbb{Z}$, which is free on one generator. Note that every free group is infinite. ▲

Refer to the literature for proofs of the next three theorems. We will not be using these results. They are stated simply to inform us of these interesting facts.

39.7 Theorem If a group G is free on A and also on B, then the sets A and B have the same number of elements; that is, any two sets of free generators of a free group have the same cardinality.

39.8 Definition If G is free on A, the number of elements in A is the **rank of the free group** G. ■

Actually, the next theorem is quite evident from Theorem 39.7.

39.9 Theorem Two free groups are isomorphic if and only if they have the same rank.

39.10 Theorem A nontrivial proper subgroup of a free group is free.

39.11 Example Let $F[\{x, y\}]$ be the free group on $\{x, y\}$. Let

$$y_k = x^k y x^{-k}$$

for $k \geq 0$. The y_k for $k \geq 0$ are free generators for the subgroup of $F[\{x, y\}]$ that they generate. This illustrates that although a subgroup of a free group is free, the rank of the subgroup may be much greater than the rank of the whole group! ▲

Homomorphisms of Free Groups

Our work in this section will be concerned primarily with homomorphisms defined on a free group. The results here are simple and elegant.

39.12 Theorem Let G be generated by $A = \{a_i \mid i \in I\}$ and let G' be any group. If a_i' for $i \in I$ are any elements in G', not necessarily distinct, then there is at most one homomorphism $\phi : G \to G'$ such that $\phi(a_i) = a_i'$. If G is free on A, then there is exactly one such homomorphism.

Proof Let ϕ be a homomorphism from G into G' such that $\phi(a_i) = a_i'$. Now by Theorem 7.6, for any $x \in G$ we have

$$x = \prod_j {a_{i_j}}^{n_j}$$

for some finite product of the generators a_i, where the a_{i_j} appearing in the product need not be distinct. Then since ϕ is a homomorphism, we must have

$$\phi(x) = \prod_j \phi\left({a_{i_j}}^{n_j}\right) = \prod_j \left({a_{i_j}}'\right)^{n_j}.$$

Thus a homomorphism is completely determined by its values on elements of a generating set. This shows that there is at most one homomorphism such that $\phi(a_i) = a_i'$.

Now suppose G is free on A; that is, $G = F[A]$. For

$$x = \prod_j {a_{i_j}}^{n_j}$$

in G, define $\psi : G \to G'$ by

$$\psi(x) = \prod_j \left({a_{i_j}}'\right)^{n_j}.$$

The map is well defined, since $F[A]$ consists precisely of reduced words; no two different formal products in $F[A]$ are equal. Since the rules for computation involving exponents in G' are formally the same as those involving exponents in G, it is clear that $\psi(xy) = \psi(x)\psi(y)$ for any elements x and y in G, so ψ is indeed a homomorphism. ◆

Perhaps we should have proved the first part of this theorem earlier, rather than having relegated it to the exercises. Note that the theorem states that *a homomorphism of a group is completely determined if we know its value on each element of a generating set*. This was Exercise 46 of Section 13. In particular, a homomorphism of a cyclic group is completely determined by its value on any single generator of the group.

39.13 Theorem Every group G' is a homomorphic image of a free group G.

Proof Let $G' = \{a_i' \mid i \in I\}$, and let $A = \{a_i \mid i \in I\}$ be a set with the same number of elements as G'. Let $G = F[A]$. Then by Theorem 39.12 there exists a homomorphism ψ mapping G into G' such that $\psi(a_i) = a_i'$. Clearly the image of G under ψ is all of G'. ◆

Another Look at Free Abelian Groups

It is important that we do not confuse the notion of a free group with the notion of a free abelian group. A free group on more than one generator is not abelian. In the preceding section, we defined a free abelian group as an abelian group that has a basis, that is, a generating set satisfying properties described in Theorem 38.1. There is another approach, via free groups, to free abelian groups. We now describe this approach.

Let $F[A]$ be the free group on the generating set A. We shall write F in place of $F[A]$ for the moment. Note that F is not abelian if A contains more than one element. Let C be the commutator subgroup of F. Then F/C is an abelian group, and it is not hard to show that F/C is free abelian with basis $\{aC \mid a \in A\}$. If aC is renamed a, we can view F/C as a free abelian group with basis A. This indicates how a free abelian group having a given set as basis can be constructed. Every free abelian group can be constructed in this fashion, up to isomorphism. That is, if G is free abelian with basis X, form the free group $F[X]$, form the factor group of $F[X]$ modulo its commutator subgroup, and we have a group isomorphic to G.

Theorems 39.7, 39.9, and 39.10 hold for free abelian groups as well as for free groups. In fact, the abelian version of Theorem 39.10 was proved for the finite rank case in Theorem 38.11. In contrast to Example 39.11 for free groups, it is true that for a free abelian group the rank of a subgroup is at most the rank of the entire group. Theorem 38.11 also showed this for the finite rank case.

EXERCISES 39

Computations

1. Find the reduced form and the inverse of the reduced form of each of the following words.
   a. $a^2b^{-1}b^3a^3c^{-1}c^4b^{-2}$ **b.** $a^2a^{-3}b^3a^4c^4c^2a^{-1}$
2. Compute the products given in parts (a) and (b) of Exercise 1 in the case that $\{a, b, c\}$ is a set of generators forming a basis for a free abelian group. Find the inverse of these products.
3. How many different homomorphisms are there of a free group of rank 2 into
   a. $\mathbb{Z}_4$? **b.** $\mathbb{Z}_6$? **c.** S_3?
4. How many different homomorphisms are there of a free group of rank 2 onto
   a. $\mathbb{Z}_4$? **b.** $\mathbb{Z}_6$? **c.** S_3?
5. How many different homomorphisms are there of a free abelian group of rank 2 into
   a. $\mathbb{Z}_4$? **b.** $\mathbb{Z}_6$? **c.** S_3?
6. How many different homomorphisms are there of a free abelian group of rank 2 onto
   a. $\mathbb{Z}_4$? **b.** $\mathbb{Z}_6$? **c.** S_3?

Concepts

In Exercises 7 and 8, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

7. A *reduced word* is one in which there are no appearances in juxtaposition of two syllables having the same letter and also no appearances of a syllable with exponent 0.

8. The *rank of a free group* is the number of elements in a set of generators for the group.

9. Take one of the instances in this section in which the phrase "It would seem obvious that" was used and discuss your reaction in that instance.

10. Mark each of the following true or false.

_____ **a.** Every proper subgroup of a free group is a free group.
_____ **b.** Every proper subgroup of every free abelian group is a free group.
_____ **c.** A homomorphic image of a free group is a free group.
_____ **d.** Every free abelian group has a basis.
_____ **e.** The free abelian groups of finite rank are precisely the finitely generated abelian groups.
_____ **f.** No free group is free.
_____ **g.** No free abelian group is free.
_____ **h.** No free abelian group of rank >1 is free.
_____ **i.** Any two free groups are isomorphic.
_____ **j.** Any two free abelian groups of the same rank are isomorphic.

Theory

11. Let G be a finitely generated abelian group with identity 0. A finite set $\{b_1, \cdots, b_n\}$, where $b_i \in G$, is a **basis for** G if $\{b_1, \cdots, b_n\}$ generates G and $\sum_{i=1}^{n} m_i b_i = 0$ if and only if each $m_i b_i = 0$, where $m_i \in \mathbb{Z}$.

a. Show that $\{2, 3\}$ is not a basis for $\mathbb{Z}_4$. Find a basis for $\mathbb{Z}_4$.
b. Show that both $\{1\}$ and $\{2, 3\}$ are bases for $\mathbb{Z}_6$. (This shows that for a finitely generated abelian group G with torsion, the number of elements in a basis may vary; that is, it need not be an *invariant* of the group G.)
c. Is a basis for a free abelian group as we defined it in Section 38 a basis in the sense in which it is used in this exercise?
d. Show that every finite abelian group has a basis $\{b_1, \cdots, b_n\}$, where the order of b_i divides the order of b_{i+1}.

In present-day expositions of algebra, a frequently used technique (particularly by the disciples of N. Bourbaki) for introducing a new algebraic entity is the following:

1. Describe algebraic properties that this algebraic entity is to possess.
2. Prove that any two algebraic entities with these properties are isomorphic, that is, that these properties characterize the entity.
3. Show that at least one such entity exists.

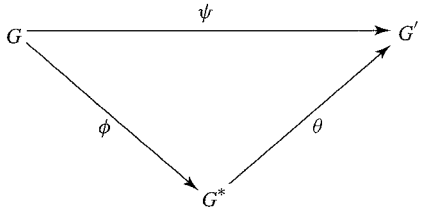
The next three exercises illustrate this technique for three algebraic entities, each of which we have met before. So that we do not give away their identities, we use fictitious names for them in the first two exercises. The last part of these first two exercises asks us to give the usual name for the entity.

12. Let G be any group. An abelian group G^* is a **blip group of** G if there exists a fixed homomorphism ϕ of G onto G^* such that each homomorphism ψ of G into an abelian group G' can be factored as $\psi = \theta\phi$, where θ is a homomorphism of G^* into G' (see Fig. 39.14).

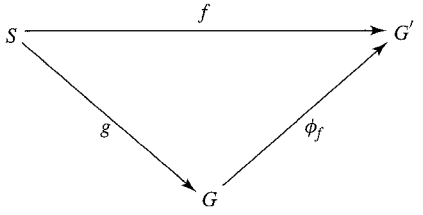
a. Show that any two blip groups of G are isomorphic. [*Hint:* Let ${G_1}^*$ and ${G_2}^*$ be two blip groups of G. Then each of the fixed homomorphisms $\phi_1 : G \to {G_1}^*$ and $\phi_2 : G \to {G_2}^*$ can be factored via the other blip group according to the definition of a blip group; that is, $\phi_1 = \theta_1\phi_2$ and $\phi_2 = \theta_2\phi_1$. Show that θ_1 is an isomorphism of ${G_2}^*$ onto ${G_1}^*$ by showing that both $\theta_1\theta_2$ and $\theta_2\theta_1$ are identity maps.]

b. Show for every group G that a blip group G^* of G exists.

c. What concept that we have introduced before corresponds to this idea of a blip group of G?



39.14 Figure



39.15 Figure

13. Let S be any set. A group G together with a fixed function $g : S \to G$ constitutes a **blop group on** S if for each group G' and map $f : S \to G'$ there exists a *unique* homomorphism ϕ_f of G into G' such that $f = \phi_f g$ (see Fig. 39.15).

a. Let S be a fixed set. Show that if both G_1, together with $g_1 : S \to G_1$, and G_2, together with $g_2 : S \to G_2$, are blop groups on S, then G_1 and G_2 are isomorphic. [*Hint:* Show that g_1 and g_2 are one-to-one maps and that g_1S and g_2S generate G_1 and G_2, respectively. Then proceed in a way analogous to that given by the hint for Exercise 12.]

b. Let S be a set. Show that a blop group on S exists. You may use any theorems of the text.

c. What concept that we have introduced before corresponds to this idea of a blop group on S?

14. Characterize a free abelian group by properties in a fashion similar to that used in Exercise 13.

SECTION 40 GROUP PRESENTATIONS

Definition

Following most of the literature on group presentations, in this section we let 1 be the identity of a group. The idea of a *group presentation* is to form a group by giving a set of generators for the group and certain equations or relations that we want the generators to satisfy. We want the group to be as free as it possibly can be on the generators, subject to these relations.

40.1 Example Suppose G has generators x and y and is *free except for the relation* $xy = yx$, which we may express as $xyx^{-1}y^{-1} = 1$. Note that the condition $xy = yx$ is exactly what is needed to make G abelian, even though $xyx^{-1}y^{-1}$ is just one of the many possible commutators of $F[\{x, y\}]$. Thus G is free abelian on two generators and is isomorphic to $F[\{x, y\}]$ modulo its commutator subgroup. This commutator subgroup of $F[\{x, y\}]$ is the smallest normal subgroup containing $xyx^{-1}y^{-1}$, since any normal subgroup

containing $xyx^{-1}y^{-1}$ gives rise to a factor group that is abelian and thus contains the commutator subgroup by Theorem 15.20. ▲

The preceding example illustrates the general situation. Let $F[A]$ be a free group and suppose that we want to form a new group as much like $F[A]$ as it can be, subject to certain equations that we want satisfied. Any equation can be written in a form in which the right-hand side is 1. Thus we can consider the equations to be $r_i = 1$ for $i \in I$, where $r_i \in F[A]$. If we require that $r_i = 1$, then we will have to have

$$x(r_i{}^n)x^{-1} = 1$$

for any $x \in F[A]$ and $n \in \mathbb{Z}$. Also any product of elements equal to 1 will again have to equal 1. Thus any finite product of the form

$$\prod_j x_j(r_{i_j}{}^{n_j})x_j^{-1},$$

where the r_{i_j} need not be distinct, will have to equal 1 in the new group. It is readily checked that the set of all these finite products is a normal subgroup R of $F[A]$. Thus any group looking as much as possible like $F[A]$, subject to the requirements $r_i = 1$, also has $r = 1$ for every $r \in R$. But $F[A]/R$ looks like $F[A]$ (remember that we multiply cosets by choosing representatives), except that R has been collapsed to form the identity 1. Hence the group we are after is (at least isomorphic to) $F[A]/R$. We can view this group as described by the generating set A and the set $\{r_i \mid i \in I\}$, which we will abbreviate $\{r_i\}$.

Historical Note

The idea of a group presentation already appears in Arthur Cayley's 1859 paper, "On the Theory of Groups as Depending on the Symbolic Equation $\theta^n = 1$. Third Part." In this article, Cayley gives a complete enumeration of the five groups of order 8, both by listing all the elements of each and by giving for each a presentation. For example, his third example is what is here called the *octic group;* Cayley notes that this group is generated by the two elements α, β with the relations $\alpha^4 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha^3$. He also shows more generally that a group of order mn is generated by α, β with the relations $\alpha^m = 1, \beta^n = 1, \alpha\beta = \beta\alpha^s$ if and only if $s^n \equiv 1 \pmod{m}$ (see Exercise 13).

In 1878, Cayley returned to the theory of groups and noted that a central problem in that theory is the determination of all groups of a given order n. In the early 1890s, Otto Hölder published several papers attempting to solve Cayley's problem. Using techniques similar to those discussed in Sections 36, 37, and 40, Hölder determined all simple groups of order up to 200 and characterized all the groups of orders p^3, pq^2, pqr, and p^4, where p, q, r are distinct prime numbers. Furthermore, he developed techniques for determining the possible structures of a group G, if one is given the structure of a normal subgroup H and the structure of the factor group G/H. Interestingly, since the notion of an abstract group was still fairly new at this time, Hölder typically began his papers with the definition of a group and also emphasized that isomorphic groups are essentially one and the same object.

40.2 Definition Let A be a set and let $\{r_i\} \subseteq F[A]$. Let R be the least normal subgroup of $F[A]$ containing the r_i. An isomorphism ϕ of $F[A]/R$ onto a group G is a **presentation of** G. The sets

A and $\{r_i\}$ give a **group presentation.** The set A is the set of **generators for the presentation** and each r_i is a **relator.** Each $r \in R$ is a **consequence of** $\{r_i\}$. An equation $r_i = 1$ is a **relation.** A **finite presentation** is one in which both A and $\{r_i\}$ are finite sets. ■

This definition may seem complicated, but it really is not. In Example 40.1, $\{x, y\}$ is our set of generators and $xyx^{-1}y^{-1}$ is the only relator. The equation $xyx^{-1}y^{-1} = 1$, or $xy = yx$, is a relation. This was an example of a finite presentation.

If a group presentation has generators x_j and relators r_i, we shall use the notations

$$(x_j : r_i) \qquad \text{or} \qquad (x_j : r_i = 1)$$

to denote the group presentation. We may refer to $F[\{x_j\}]/R$ as *the group with presentation* $(x_j : r_i)$.

Isomorphic Presentations

40.3 Example Consider the group presentation with

$$A = \{a\} \qquad \text{and} \qquad \{r_i\} = \{a^6\},$$

that is, the presentation

$$(a : a^6 = 1).$$

This group defined by one generator a, with the relation $a^6 = 1$, is isomorphic to $\mathbb{Z}_6$.

Now consider the group defined by two generators a and b, with $a^2 = 1$, $b^3 = 1$, and $ab = ba$, that is, the group with presentation

$$(a, b : a^2, b^3, aba^{-1}b^{-1}).$$

The condition $a^2 = 1$ gives $a^{-1} = a$. Also $b^3 = 1$ gives $b^{-1} = b^2$. Thus every element in this group can be written as a product of nonnegative powers of a and b. The relation $aba^{-1}b^{-1} = 1$, that is, $ab = ba$, allows us to write first all the factors involving a and then the factors involving b. Hence every element of the group is equal to some $a^m b^n$. But then $a^2 = 1$ and $b^3 = 1$ show that there are just six distinct elements,

$$1, b, b^2, a, ab, ab^2.$$

Therefore this presentation also gives a group of order 6 that is abelian, and by the Fundamental Theorem 11.12, it must again be cyclic and isomorphic to $\mathbb{Z}_6$. ▲

The preceding example illustrates that different presentations may give isomorphic groups. When this happens, we have **isomorphic presentations.** To determine whether two presentations are isomorphic may be very hard. It has been shown (see Rabin [22]) that a number of such problems connected with this theory are not generally solvable; that is, there is no *routine* and well-defined way of discovering a solution in all cases. These unsolvable problems include the problem of deciding whether two presentations are isomorphic, whether a group given by a presentation is finite, free, abelian, or trivial, and the famous *word problem* of determining whether a given word w is a consequence of a given set of relations $\{r_i\}$.

The importance of this material is indicated by our Theorem 39.13, which guarantees that *every group has a presentation.*

40.4 Example Let us show that

$$(x, y : y^2x = y, yx^2y = x)$$

is a presentation of the trivial group of one element. We need only show that x and y are consequences of the relators y^2xy^{-1} and yx^2yx^{-1}, or that $x = 1$ and $y = 1$ can be deduced from $y^2x = y$ and $yx^2y = x$. We illustrate both techniques.

As a consequence of y^2xy^{-1}, we get yx upon conjugation by y^{-1}. From yx we deduce $x^{-1}y^{-1}$, and then $(x^{-1}y^{-1})(yx^2yx^{-1})$ gives xyx^{-1}. Conjugating xyx^{-1} by x^{-1}, we get y. From y we get y^{-1}, and $y^{-1}(yx)$ is x.

Working with relations instead of relators, from $y^2x = y$ we deduce $yx = 1$ upon multiplication by y^{-1} on the left. Then substituting $yx = 1$ into $yx^2y = x$, that is, $(yx)(xy) = x$, we get $xy = x$. Then multiplying by x^{-1} on the left, we have $y = 1$. Substituting this in $yx = 1$, we get $x = 1$.

Both techniques amount to the same work, but it somehow seems more natural to most of us to work with relations. ▲

Applications

We conclude this chapter with two applications.

40.5 Example Let us determine all groups of order 10 up to isomorphism. We know from the Fundamental Theorem 11.12 that every abelian group of order 10 is isomorphic to $\mathbb{Z}_{10}$. Suppose that G is nonabelian of order 10. By Sylow theory, G contains a normal subgroup H of order 5, and H must be cyclic. Let a be a generator of H. Then G/H is of order 2 and thus isomorphic to Z_2. If $b \in G$ and $b \notin H$, we must then have $b^2 \in H$. Since every element of H except 1 has order 5, if b^2 were not equal to 1, then b^2 would have order 5, so b would have order 10. This would mean that G would be cyclic, contradicting our assumption that G is not abelian. Thus $b^2 = 1$. Finally, since H is a normal subgroup of G, $bHb^{-1} = H$, so in particular, $bab^{-1} \in H$. Since conjugation by b is an automorphism of H, bab^{-1} must be another element of H of order 5, hence bab^{-1} equals a, a^2, a^3, or a^4. But $bab^{-1} = a$ would give $ba = ab$, and then G would be abelian, since a and b generate G. Thus the possibilities for presentations of G are:

1. $(a, b : a^5 = 1, b^2 = 1, ba = a^2b)$,
2. $(a, b : a^5 = 1, b^2 = 1, ba = a^3b)$,
3. $(a, b : a^5 = 1, b^2 = 1, ba = a^4b)$.

Note that all three of these presentations can give groups of order at most 10, since the last relation $ba = a^ib$ enables us to express every product of a's and b's in G in the form a^sb^t. Then $a^5 = 1$ and $b^2 = 1$ show that the set

$$S = \{a^0b^0, a^1b^0, a^2b^0, a^3b^0, a^4b^0, a^0b^1, a^1b^1, a^2b^1, a^3b^1, a^4b^1\}$$

includes all elements of G.

It is not yet clear that all these elements in S are distinct, so that we have in all three cases a group of order 10. For example, the group presentation

$$(a, b : a^5 = 1, b^2 = 1, ba = a^2b)$$

gives a group in which, using the associative law, we have

$$\begin{aligned} a &= b^2a = (bb)a = b(ba) = b(a^2b) = (ba)(ab) \\ &= (a^2b)(ab) = a^2(ba)b = a^2(a^2b)b = a^4b^2 = a^4 \end{aligned}$$

Thus in this group, $a = a^4$, so $a^3 = 1$, which, together with $a^5 = 1$, yields $a^2 = 1$. But $a^2 = 1$, together with $a^3 = 1$, means that $a = 1$. Hence every element in the group with presentation

$$(a, b : a^5 = 1, b^2 = 1, ba = a^2b)$$

is equal to either 1 or b; that is, this group is isomorphic to $\mathbb{Z}_2$. A similar study of

$$(bb)a = b(ba)$$

for

$$(a, b : a^5 = 1, b^2 = 1, ba = a^3b)$$

shows that $a = a^4$ again, so this also yields a group isomorphic to $\mathbb{Z}_2$.

This leaves just

$$(a, b : a^5 = 1, b^2 = 1, ba = a^4b)$$

as a candidate for a nonabelian group of order 10. In this case, it can be shown that all elements of S are distinct, so this presentation does give a nonabelian group G of order 10. How can we show that all elements in S represent distinct elements of G? The easy way is to observe that we know that there is at least one nonabelian group of order 10, the dihedral group D_5. Since G is the only remaining candidate, we must have $G \simeq D_5$. Another attack is as follows. Let us try to make S into a group by defining $(a^sb^t)(a^ub^v)$ to be a^xb^y, where x is the remainder of $s + u(4^t)$ when divided by 5, and y is the remainder of $t + v$ when divided by 2, in the sense of the division algorithm (Theorem 6.3). In other words, we use the relation $ba = a^4b$ as a guide in defining the product $(a^sb^t)(a^ub^v)$ of two elements of S. We see that a^0b^0 acts as identity, and that given a^ub^v, we can determine t and s successively by letting

$$t \equiv -v \pmod 2$$

and then

$$s \equiv -u(4^t) \pmod 5,$$

giving a^sb^t, which is a left inverse for a^ub^v. We will then have a group structure on S if and only if the associative law holds. Exercise 13 asks us to carry out the straight-forward computation for the associative law and to discover a condition for S to be a group under such a definition of multiplication. The criterion of the exercise in this case amounts to the valid congruence

$$4^2 \equiv 1 \pmod 5.$$

Thus we do get a group of order 10. Note that

$$2^2 \not\equiv 1 \pmod 5$$

and

$$3^2 \not\equiv 1 \pmod 5,$$

so Exercise 13 also shows that

$$(a, b : a^5 = 1, b^2 = 1, ba = a^2b)$$

and

$$(a, b : a^5 = 1, b^2 = 1, ba = a^3b)$$

do not give groups of order 10. ▲

40.6 Example Let us determine all groups of order 8 up to isomorphism. We know the three abelian ones:

$$\mathbb{Z}_8, \qquad \mathbb{Z}_2 \times \mathbb{Z}_4, \qquad \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Using generators and relations, we shall give presentations of the nonabelian groups.

Let G be nonabelian of order 8. Since G is nonabelian, it has no elements of order 8, so each element but the identity is of order either 2 or 4. If every element were of order 2, then for $a, b \in G$, we would have $(ab)^2 = 1$, that is, $abab = 1$. Then since $a^2 = 1$ and $b^2 = 1$ also, we would have

$$ba = a^2bab^2 = a(ab)^2b = ab,$$

contrary to our assumption that G is not abelian. Thus G must have an element of order 4.

Let $\langle a \rangle$ be a subgroup of G of order 4. If $b \notin \langle a \rangle$, the cosets $\langle a \rangle$ and $b\langle a \rangle$ exhaust all of G. Hence a and b are generators for G and $a^4 = 1$. Since $\langle a \rangle$ is normal in G (by Sylow theory, or because it is of index 2), $G/\langle a \rangle$ is isomorphic to $\mathbb{Z}_2$ and we have $b^2 \in \langle a \rangle$. If $b^2 = a$ or $b^2 = a^3$, then b would be of order 8. Hence $b^2 = 1$ or $b^2 = a^2$. Finally, since $\langle a \rangle$ is normal, we have $bab^{-1} \in \langle a \rangle$, and since $b\langle a \rangle b^{-1}$ is a subgroup conjugate to $\langle a \rangle$ and hence isomorphic to $\langle a \rangle$, we see that bab^{-1} must be an element of order 4. Thus $bab^{-1} = a$ or $bab^{-1} = a^3$. If bab^{-1} were equal to a, then ba would equal ab, which would make G abelian. Hence $bab^{-1} = a^3$, so $ba = a^3b$. Thus we have two possibilities for G, namely,

$$G_1 : (a, b : a^4 = 1, b^2 = 1, ba = a^3b)$$

and

$$G_2 : (a, b : a^4 = 1, b^2 = a^2, ba = a^3b).$$

Note that $a^{-1} = a^3$, and that b^{-1} is b in G_1 and b^3 in G_2. These facts, along with the relation $ba = a^3b$, enable us to express every element in G_i in the form a^mb^n, as in Examples 40.3 and 40.5. Since $a^4 = 1$ and either $b^2 = 1$ or $b^2 = a^2$, the possible elements in each group are

$$1, \quad a, \quad a^2, \quad a^3, \quad b, \quad ab, \quad a^2b, \quad a^3b.$$

Thus G_1 and G_2 each have order at most 8. That G_1 is a group of order 8 can be seen from Exercise 13. An argument similar to that used in Exercise 13 shows that G_2 has order 8 also.

Since $ba = a^3b \neq ab$, we see that both G_1 and G_2 are nonabelian. That the two groups are not isomorphic follows from the fact that a computation shows that G_1 has only two elements of order 4, namely, a and a^3. On the other hand, in G_2 all elements but 1 and a^2 are of order 4. We leave the computations of the tables for these groups to Exercise 3. To illustrate suppose we wish to compute $(a^2b)(a^3b)$. Using $ba = a^3b$ repeatedly, we get

$$(a^2b)(a^3b) = a^2(ba)a^2b = a^5(ba)ab = a^8(ba)b = a^{11}b^2.$$

Then for G_1, we have

$$a^{11}b^2 = a^{11} = a^3,$$

but if we are in G_2, we get

$$a^{11}b^2 = a^{13} = a.$$

The group G_1 is the **octic group** and is isomorphic to our old friend, the group D_4 of symmetries of the square. The group G_2 is the **quaternion group;** it is isomorphic to the multiplicative group $\{1, -1, i, -i, j, -j, k, -k\}$ of quaternions. Quaternions were discussed in Section 24. ▲

EXERCISES 40

Computations

1. Give a presentation of $\mathbb{Z}_4$ involving one generator; involving two generators; involving three generators.
2. Give a presentation of S_3 involving three generators.
3. Give the tables for both the octic group

$$(a, b : a^4 = 1, b^2 = 1, ba = a^3b)$$

and the quaternion group

$$(a, b : a^4 = 1, b^2 = a^2, ba = a^3b).$$

In both cases, write the elements in the order $1, a, a^2, a^3, b, ab, a^2b, a^3b$. (Note that we do not have to compute *every* product. We know that these presentations give groups of order 8, and once we have computed enough products the rest are forced so that each row and each column of the table has each element exactly once.)

4. Determine all groups of order 14 up to isomorphism. [*Hint:* Follow the outline of Example 40.5 and use Exercise 13, part (b).]
5. Determine all groups of order 21 up to isomorphism. [*Hint:* Follow the outline of Example 40.5 and use Exercise 13, part (b). It may seem that there are two presentations giving nonabelian groups. Show that they are isomorphic.]

Concepts

In Exercises 6 and 7, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

6. A *consequence* of the set of relators is any finite product of relators raised to powers.

7. Two group presentations are *isomorphic* if and only if there is a one-to-one correspondence of the generators of the first presentation with the generators of the second that yields, by renaming generators, a one-to-one correspondence of the relators of the first presentation with those of the second.

8. Mark each of the following true or false.

_____ **a.** Every group has a presentation.
_____ **b.** Every group has many different presentations.
_____ **c.** Every group has two presentations that are not isomorphic.
_____ **d.** Every group has a finite presentation.
_____ **e.** Every group with a finite presentation is of finite order.
_____ **f.** Every cyclic group has a presentation with just one generator.
_____ **g.** Every conjugate of a relator is a consequence of the relator.
_____ **h.** Two presentations with the same number of generators are always isomorphic.
_____ **i.** In a presentation of an abelian group, the set of consequences of the relators contains the commutator subgroup of the free group on the generators.
_____ **j.** Every presentation of a free group has 1 as the only relator.

Theory

9. Use the methods of this section and Exercise 13, part (b), to show that there are no nonabelian groups of order 15. (See also Example 37.10).

10. Show, using Exercise 13, that

$$(a, b : a^3 = 1, b^2 = 1, ba = a^2b)$$

gives a group of order 6. Show that it is nonabelian.

11. Show that the presentation

$$(a, b : a^3 = 1, b^2 = 1, ba = a^2b)$$

of Exercise 10 gives (up to isomorphism) the only nonabelian group of order 6, and hence gives a group isomorphic to S_3.

12. We showed in Example 15.6 that A_4 has no subgroup of order 6. The preceding exercise shows that such a subgroup of A_4 would have to be isomorphic to either $\mathbb{Z}_6$ or S_3. Show again that this is impossible by considering orders of elements.

13. Let

$$S = \{a^i b^j \mid 0 \leq i < m, 0 \leq j < n\},$$

that is, S consists of all formal products $a^i b^j$ starting with $a^0 b^0$ and ending with $a^{m-1} b^{n-1}$. Let r be a positive integer, and define multiplication on S by

$$(a^s b^t)(a^u b^v) = a^x b^y,$$

where x is the remainder of $s + u(r^t)$ when divided by m, and y is the remainder of $t + v$ when divided by n, in the sense of the division algorithm (Theorem 6.3).

a. Show that a necessary and sufficient condition for the associative law to hold and for S to be a group under this multiplication is that $r^n \equiv 1 \pmod{m}$.

b. Deduce from part (a) that the group presentation

$$(a, b : a^m = 1, b^n = 1, ba = a^r b)$$

gives a group of order mn if and only if $r^n \equiv 1 \pmod{m}$. (See the Historical Note on page xxx.)

14. Show that if $n = pq$, with p and q primes and $q > p$ and $q \equiv 1 \pmod{p}$, then there is exactly one nonabelian group (up to isomorphism) of order n. Recall that the $q - 1$ nonzero elements of $\mathbb{Z}_q$ form a cyclic group $\mathbb{Z}_q^*$ under multiplication modulo q. [*Hint:* The solutions of $x^p \equiv 1 \pmod{q}$ form a cyclic subgroup of $\mathbb{Z}_q^*$ with elements $1, r, r^2, \cdots, r^{p-1}$. In the group with presentation $(a, b : a^q = 1, b^p = 1, ba = a^r b)$, we have $bab^{-1} = a^r$, so $b^j a b^{-j} = a^{(r^j)}$. Thus, since b^j generates $\langle b \rangle$ for $j = 1, \cdots, p - 1$, this presentation is isomorphic to

$$(a, b^j : a^q = 1, (b^j)^p = 1, (b^j)a = a^{(r^j)}(b^j)),$$

so all the presentations $(a, b : a^q = 1, b^p = 1, ba = a^{(r^j)}b)$ are isomorphic.]